

Biological Data Analysis

(a.k.a. Computermethoden der Bioinformatik)

Carl Herrmann
Health Data Science Unit
Medizinische Fakultät

Maiwen Caudron-Herger
DKFZ



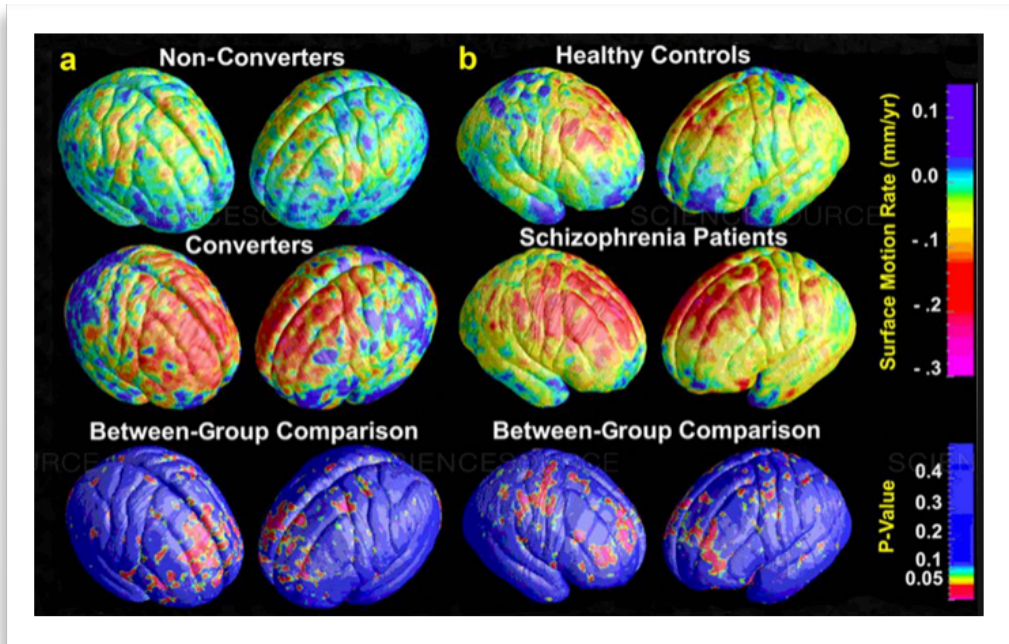
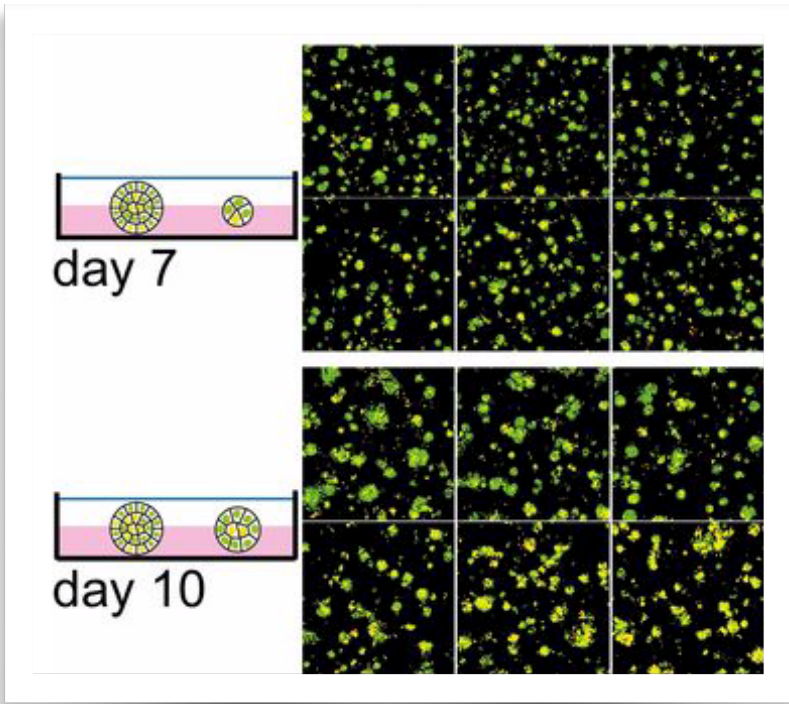
UNIVERSITÄT
HEIDELBERG
ZUKUNFT
SEIT 1386

0. Basics about data types

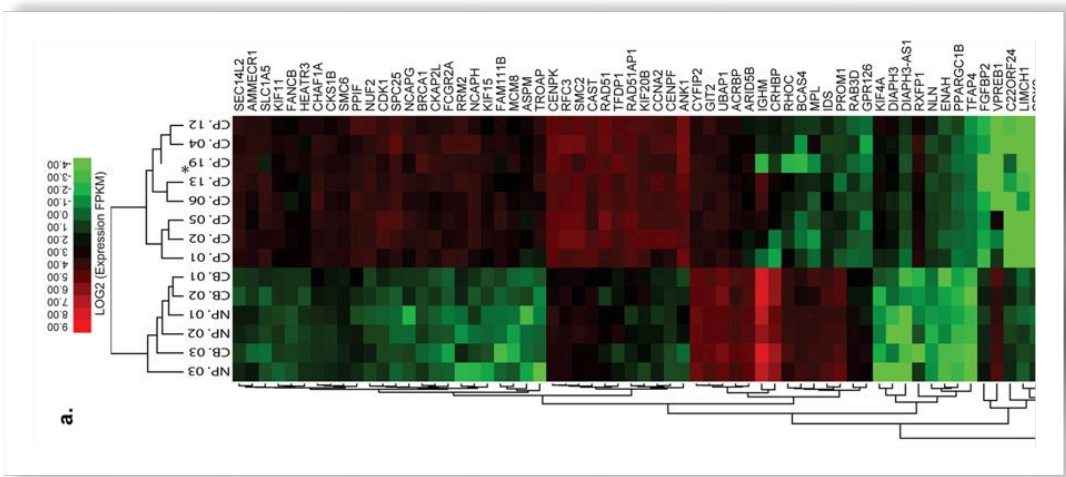
Sources of biological data

Biomedical images

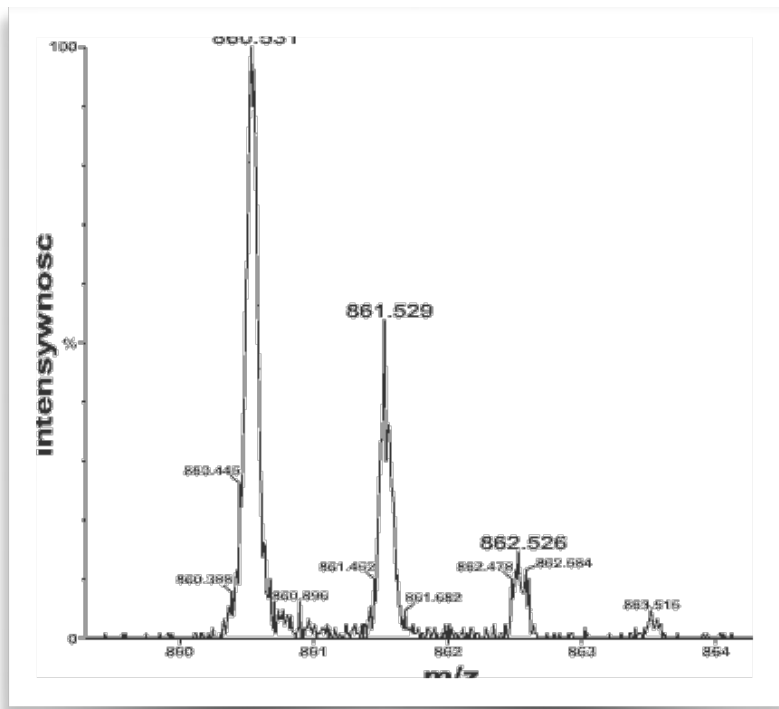
Drug screens



Genomics



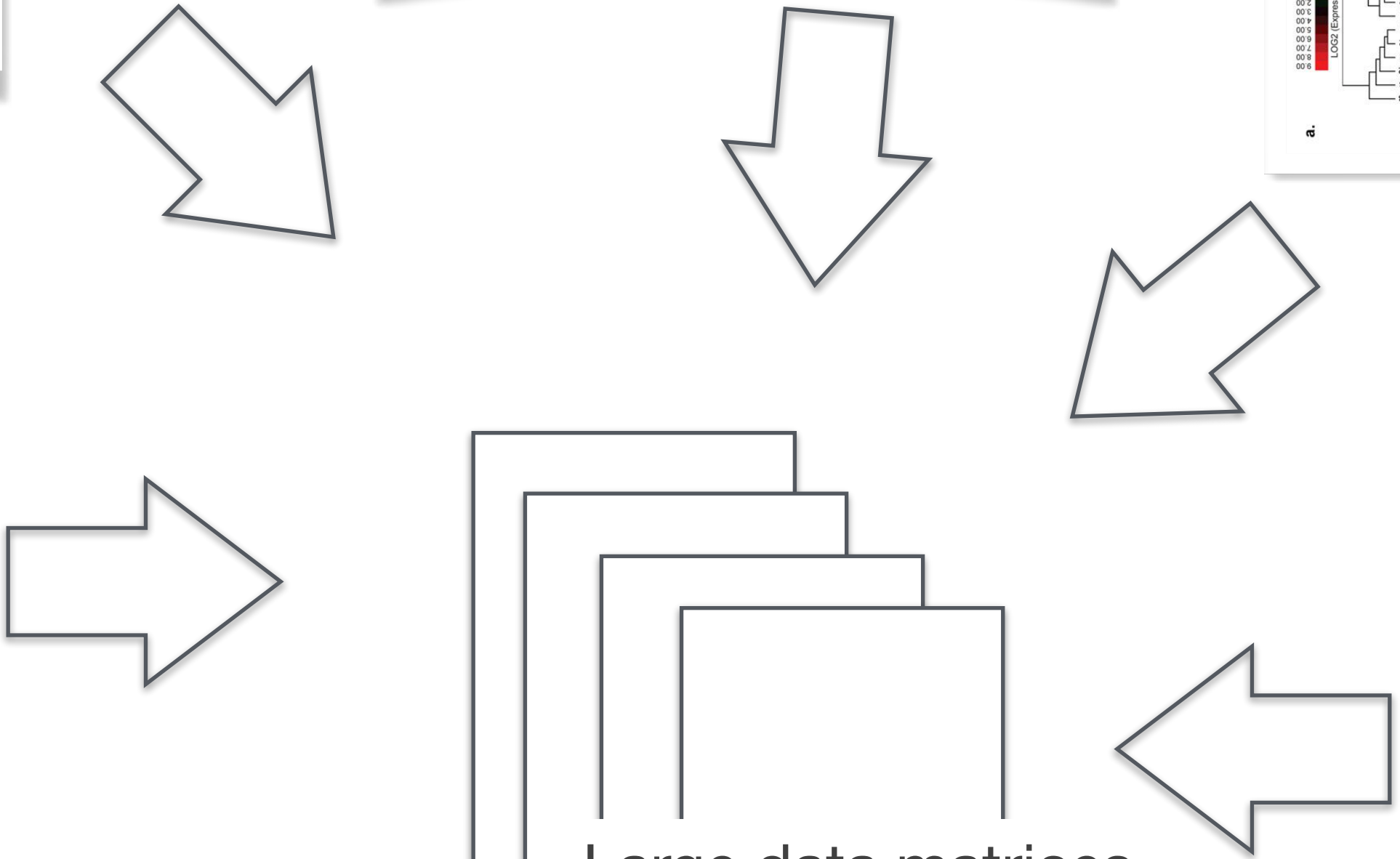
Proteomics



Patient data

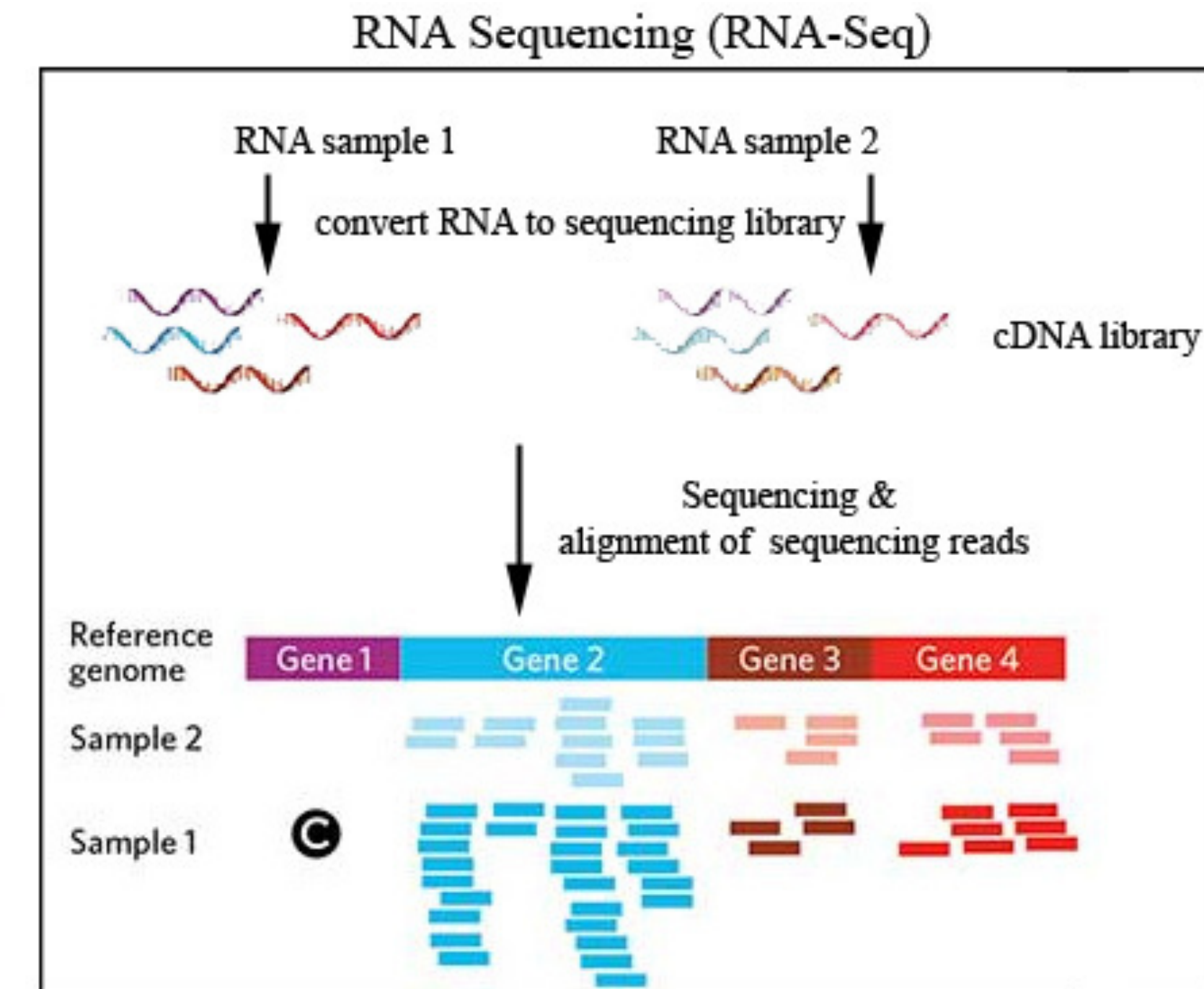


Large data matrices



Example : gene expression

- mRNA molecules are retro-transcribed into cDNA
- cDNA libraries are sequenced into short fragments (~ 100 bp)
- Fragments are aligned to the genome
- Normalized number of fragments per gene is a measure of the gene expression
- examples of normalizations: reads per kilobase per million aligned reads (RPKM)



Example : gene expression

Genes
(n ~30.000)

Patients
(n ~ 100 - 1000)

Description	GTEX-1117F-0226-SM-5GZZ7	GTEX-111CU-1826-SM-5GZY N	GTEX-111FC-0226-SM-5N9B 8	GTEX-111VG-2326-SM-5N9BK	GTEX-111YS-2426-SM-5GZZQ	GTEX-1122O-2026-SM-5NQ91	GTEX-1128S-2126-SM-5H12U	GTEX-113IC-0226-SM-5HL5C
DDX11L1	3	4	1	1	0	2	1	3
WASH7P	616	395	826	364	301	419	340	451
MIR1302-11	2	1	1	0	1	0	2	3
FAM138A	1	0	1	1	0	0	2	3
OR4G4P	0	0	0	0	0	0	2	1
OR4G11P	0	2	2	0	0	1	0	0
OR4F5	0	0	0	0	2	0	0	0
RP11-34P13.7	8	3	12	12	2	4	10	9
CICP27	11	29	9	18	5	5	7	4

Example : gene expression

Gene expression values

Gene

Patienten

genes	gene_1	gene_2	gene_3	gene_4
patient_1	4.14391122151671	2.97523610717862	-3.46753471324685	-0.0542174928459582
patient_2	3.09824631414336	1.20443006241933	-2.00101678051065	0.886366385234324
patient_3	0.613539225627942	4.3447882784591	2.58908155510034	1.3073011508442
patient_4	4.01373315182408	3.85312401281436	3.31290558096149	3.59288980826203
patient_5	0.383657955269923	2.99916338009409	-0.519502934359013	-1.68381158067256
patient_6	3.48703239461242	1.05441291342982	2.06695160803053	0.825554440360175
patient_7	0.515634555377573	0.96825404034163	-4.24026922022054	-2.15459287415868
patient_8	0.833008182831021	1.66882594396195	3.54523865807237	3.91542374367522
patient_9	3.27748316735109	3.44774731855558	3.87916459892164	2.37301081167968
patient_10	2.4007858903261	-1.25740300340426	2.39681703044223	2.47361107715352
patient_11	2.52460225221238	0.705443998616781	0.305480184099154	4.1439933583416
patient_12	1.14585639287521	0.876609151235227	1.06556812351244	0.857585380566236
patient_13	0.912240962281692	1.43642915421264	2.21361502482881	0.191184996769756

Tables are oriented most of the time such that # rows > # columns

Example: clinical data

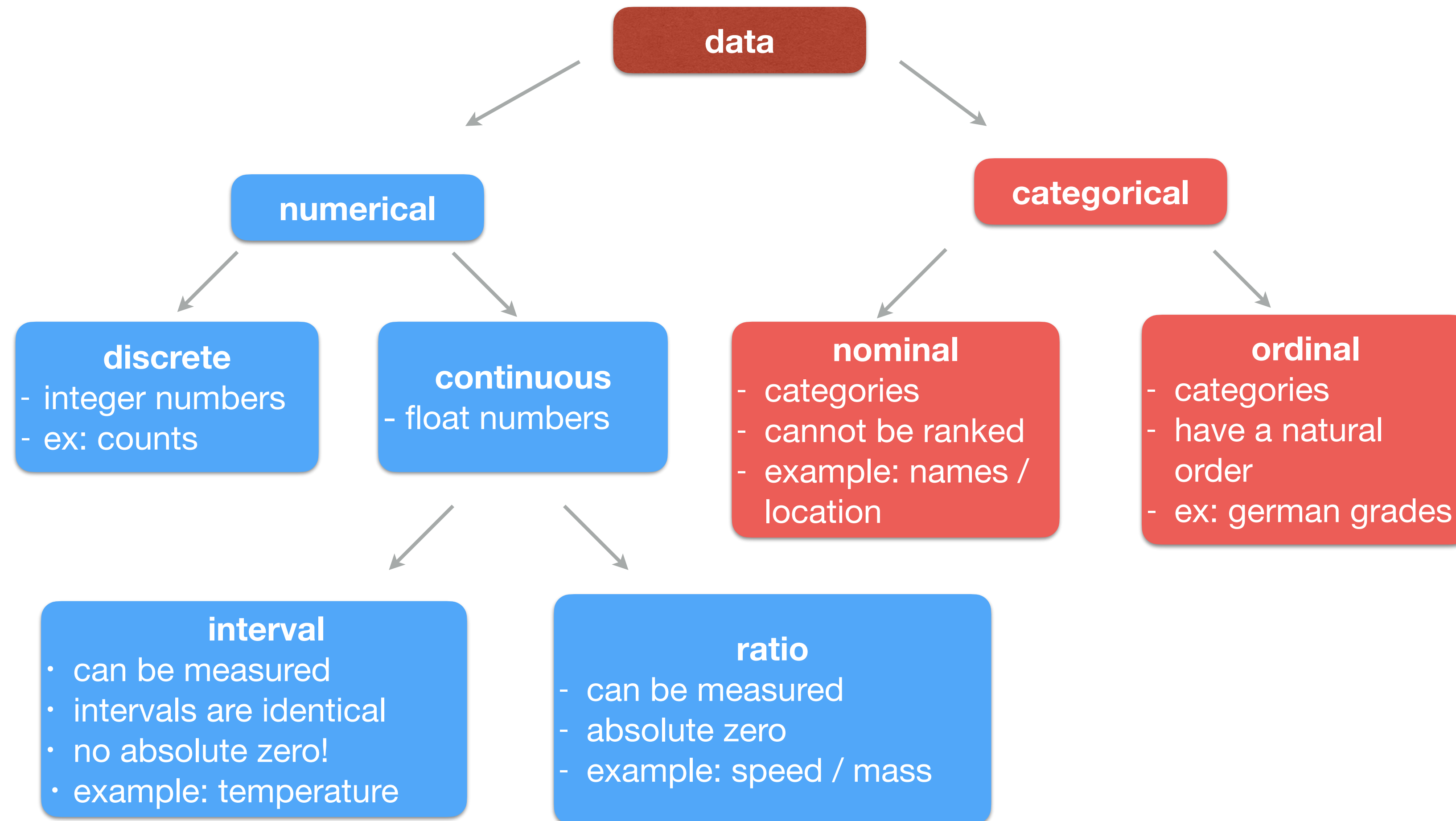
cohort of diabetes patients

id	chol	stab.glu	hdl	ratio	glyhb	location	age	gender	height	weight	frame	bp.1s	bp.1d	bp.2s	bp.2d	waist	hip	time.ppn
1000	203	82	56	3,60	4,31	Buckingham	46	female	62	121	medium	118	59	NA	NA	29	38	720
1001	165	97	24	6,90	4,44	Buckingham	29	female	64	218	large	112	68	NA	NA	46	48	360
1002	228	92	37	6,20	4,64	Buckingham	58	female	61	256	large	190	92	185	92	49	57	180
1003	78	93	12	6,50	4,63	Buckingham	67	male	67	119	large	110	50	NA	NA	33	38	480
1005	249	90	28	8,90	7,72	Buckingham	64	male	68	183	medium	138	80	NA	NA	44	41	300
1008	248	94	69	3,60	4,81	Buckingham	34	male	71	190	large	132	86	NA	NA	36	42	195
1011	195	92	41	4,80	4,84	Buckingham	30	male	69	191	medium	161	112	161	112	46	49	720
1015	227	75	44	5,20	3,94	Buckingham	37	male	59	170	medium	NA	NA	NA	NA	34	39	1020
1016	177	87	49	3,60	4,84	Buckingham	45	male	69	166	large	160	80	128	86	34	40	300
1022	263	89	40	6,60	5,78	Buckingham	55	female	63	202	small	108	72	NA	NA	45	50	240
1024	242	82	54	4,50	4,77	Louisa	60	female	65	156	medium	130	90	130	90	39	45	300
1029	215	128	34	6,30	4,97	Louisa	38	female	58	195	medium	102	68	NA	NA	42	50	90
1030	238	75	36	6,60	4,47	Louisa	27	female	60	170	medium	130	80	NA	NA	35	41	720
1031	183	79	46	4,00	4,59	Louisa	40	female	59	165	medium	NA	NA	NA	NA	37	43	60
1035	191	76	30	6,40	4,67	Louisa	36	male	69	183	medium	100	66	NA	NA	36	40	225
1036	213	83	47	4,50	3,41	Louisa	33	female	65	157	medium	130	90	120	96	37	41	240
1037	255	78	38	6,70	4,33	Louisa	50	female	65	183	medium	130	100	NA	NA	37	43	180

Different types of data

Variable	Explanation	Unit	Type
chol	total cholesterol		numerical
stab.glu	Stabilized Glucose		numerical
hdl	High Density Lipoprotein		numerical
ratio	Cholesterol/HDL Ratio		numerical
glyhb	Glycosolated Hemoglobin		numerical
location			categorical
age			numerical
gender			categorical
height		inches	numerical
weight		pounds	numerical
frame			categorical
bp.1s	systolic blood pressure		numerical
bp.1d	diastolic blood pressure		numerical
bp.2s	systolic blood pressure		numerical
bp.2d	diastolic blood pressure		numerical
waist		inches	numerical
hip		inches	numerical
time.ppn	Time since last meal	minutes	numerical

Different types of data



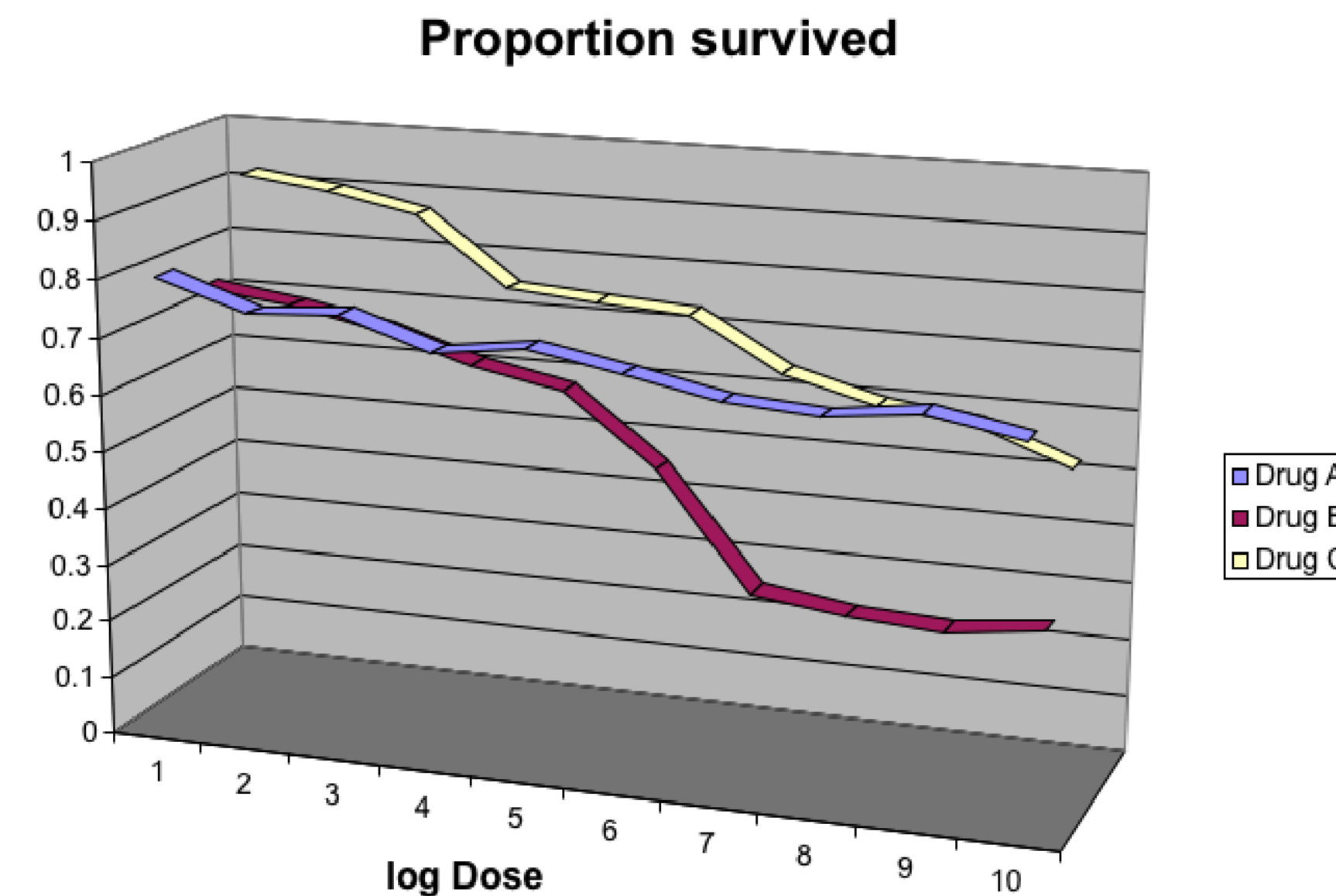
Different types of data

Variable	Explanation	Unit	Type	Subtype
chol	total cholesterol		numerical	ratio
stab.glu	Stabilized Glucose		numerical	ratio
hdl	High Density Lipoprotein		numerical	ratio
ratio	Cholesterol/HDL Ratio		numerical	ratio
glyhb	Glycosolated Hemoglobin		numerical	ratio
location			categorical	nominal
age			numerical	ratio
gender			categorical	nominal
height		inches	numerical	ratio
weight		pounds	numerical	ratio
frame			categorical	ordinal (small/medium/large)
bp.1s	systolic blood pressure		numerical	ratio
bp.1d	diastolic blood pressure		numerical	ratio
bp.2s	systolic blood pressure		numerical	ratio
bp.2d	diastolic blood pressure		numerical	ratio
waist		inches	numerical	ratio
hip		inches	numerical	ratio
time.ppn	Time since last meal	minutes	numerical	ratio

1. Data visualization - Plots

Graphical representation

- Appropriate graphical representation depends on the type of data
 - categorical
 - counts
 - continuous data
- Aim of good data graphics: display data accurately and clearly (Karl Broman https://www.biostat.wisc.edu/~kbroman/topten_worstgraphs/)
- Bad practice:
 - as little information as possible
 - make things obscure through inappropriate graphics
 - pseudo 3D
 - poor scales



Example of bad plot...

What is wrong with this plot?

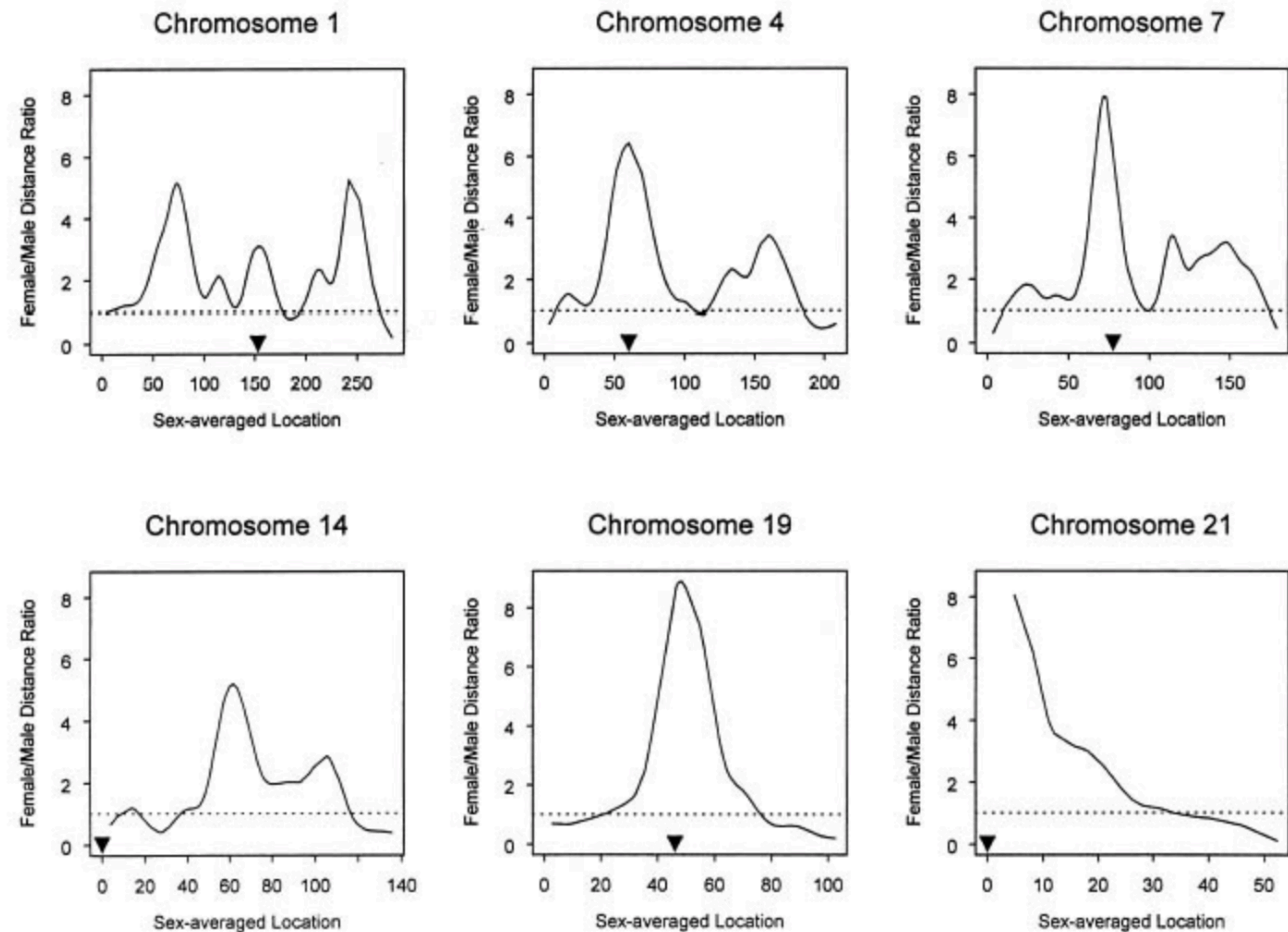
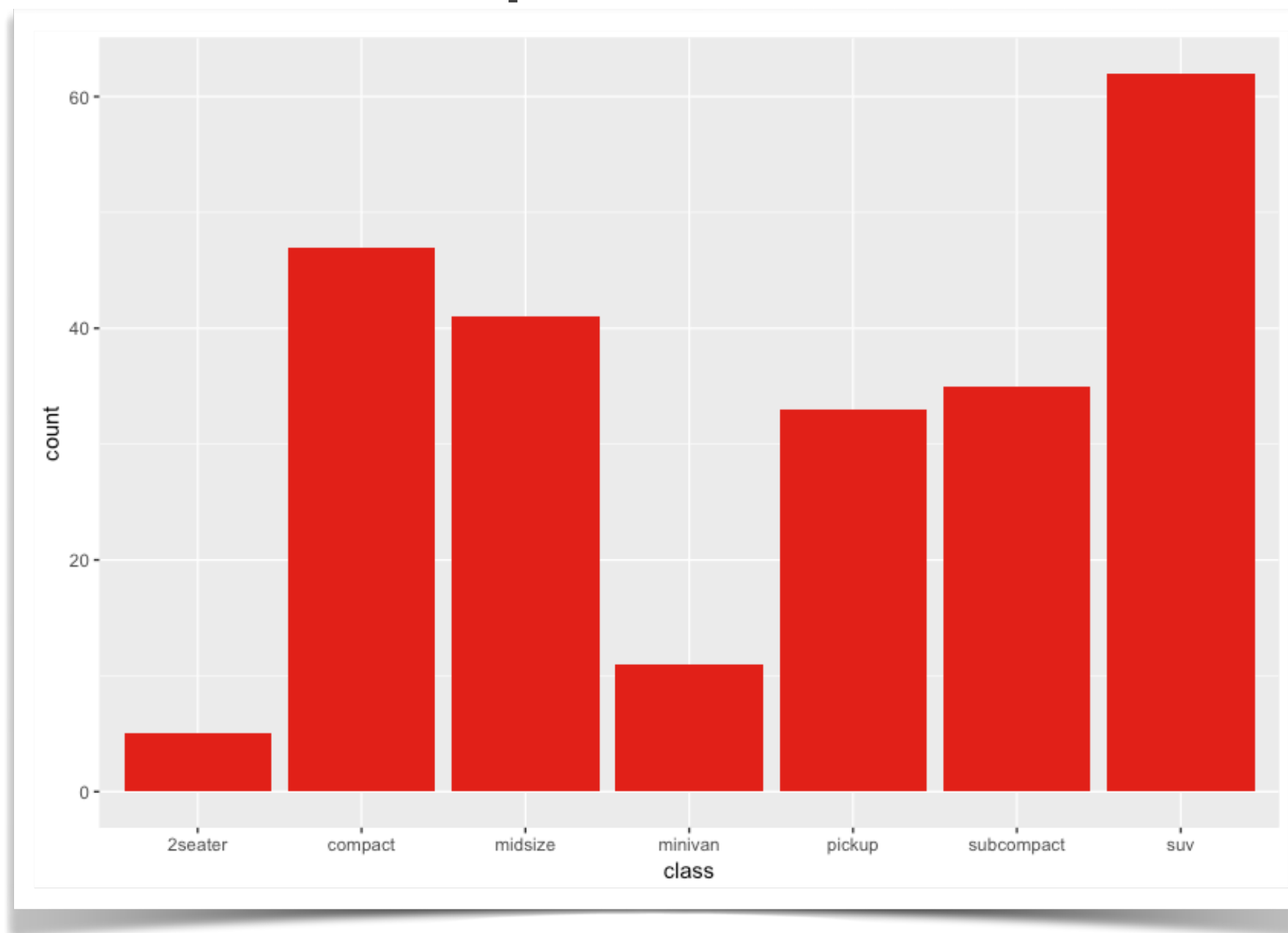


Figure 1 Plots of the female:male genetic-distance ratio against sex-averaged genetic location (in cM) along six selected chromosomes. Approximate locations of the centromeres are indicated by the triangles. The dashed lines correspond to equal female and male distances.

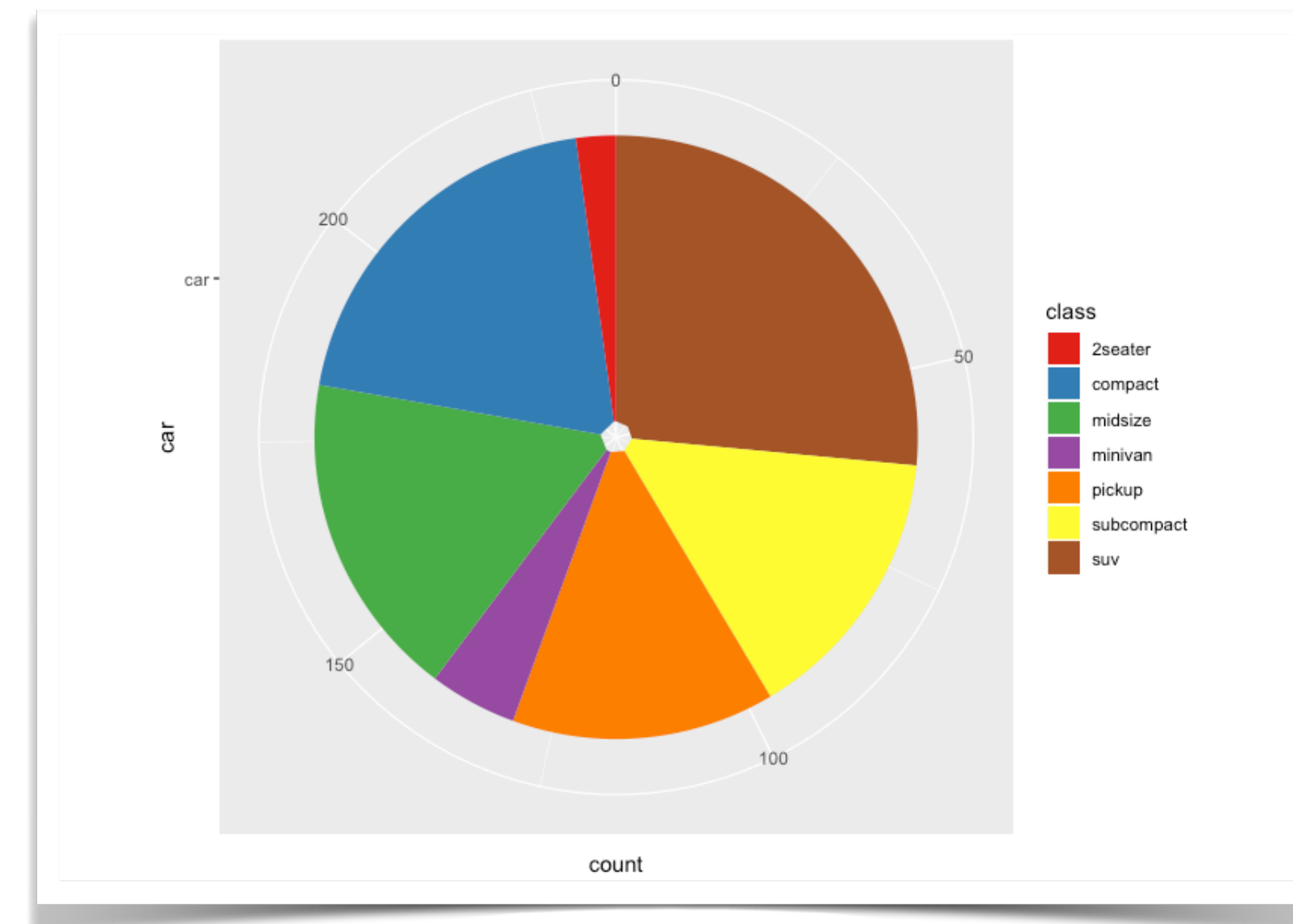
Categorical data - barplot

- How many instances in each category?
- Only meaningful measure: **MODE** (= category with highest counts)
- Possible plots: barplots; piecharts

Barplot



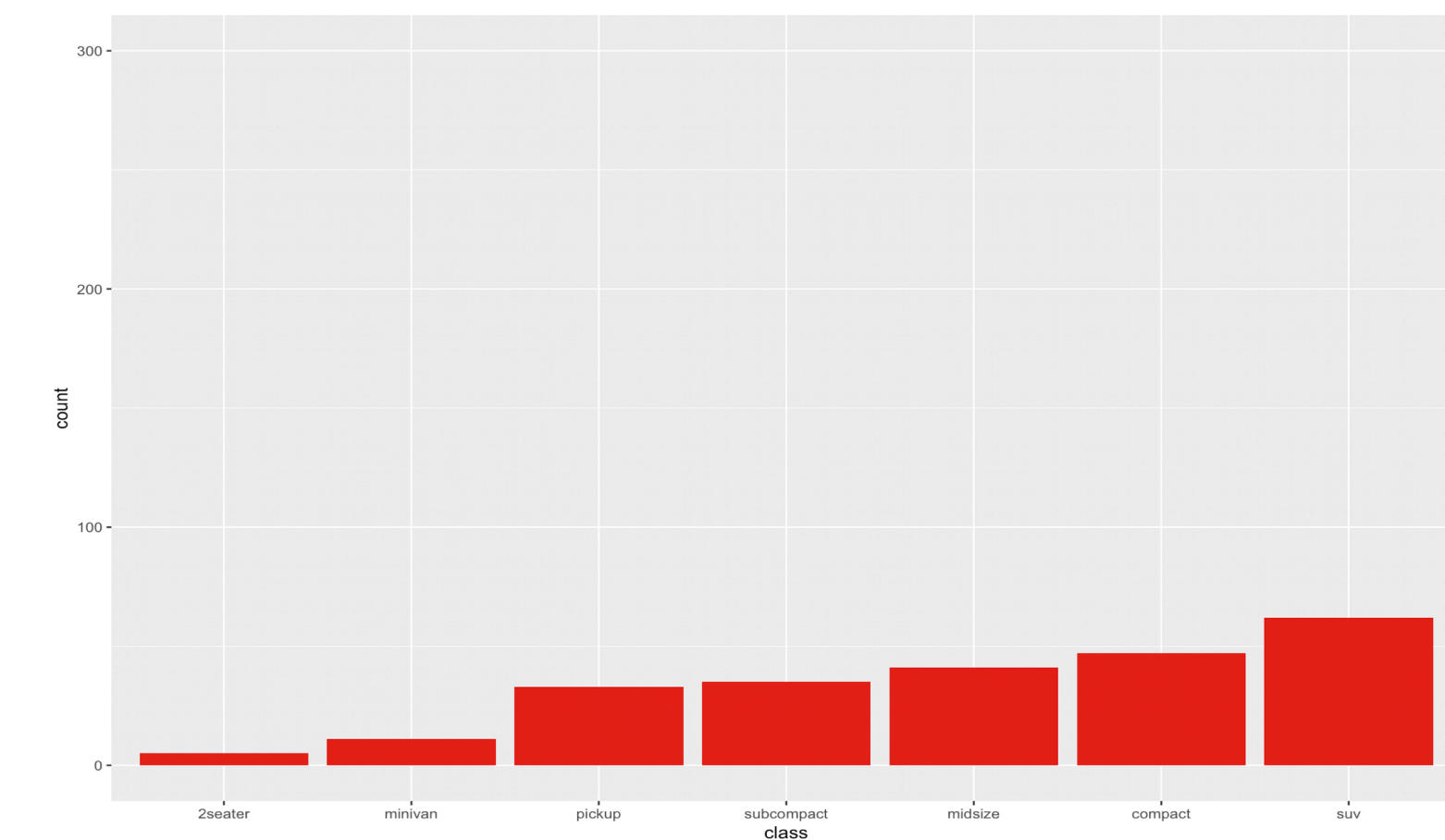
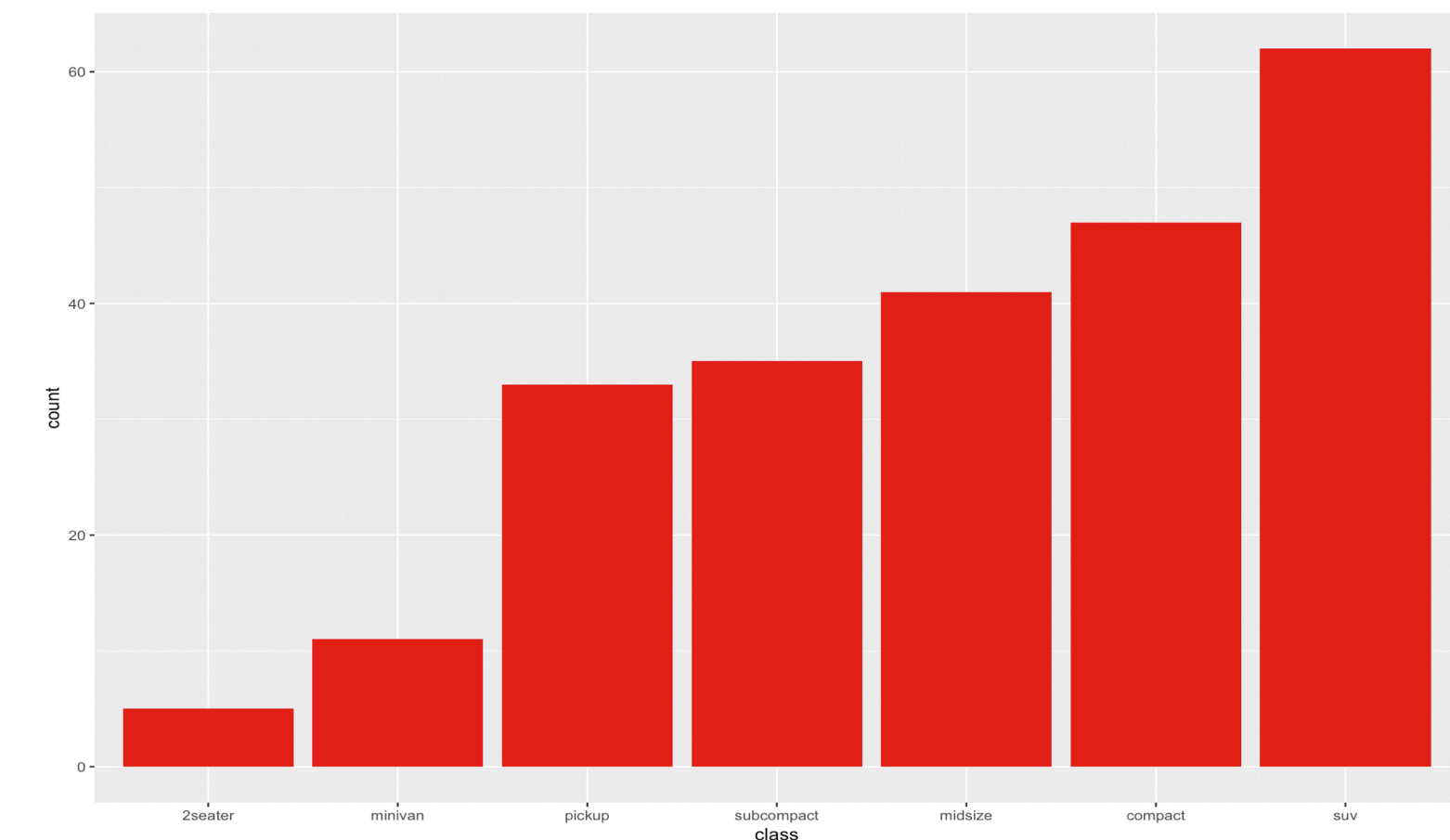
Piechart



Avoid piecharts: areas are less intuitive than length!

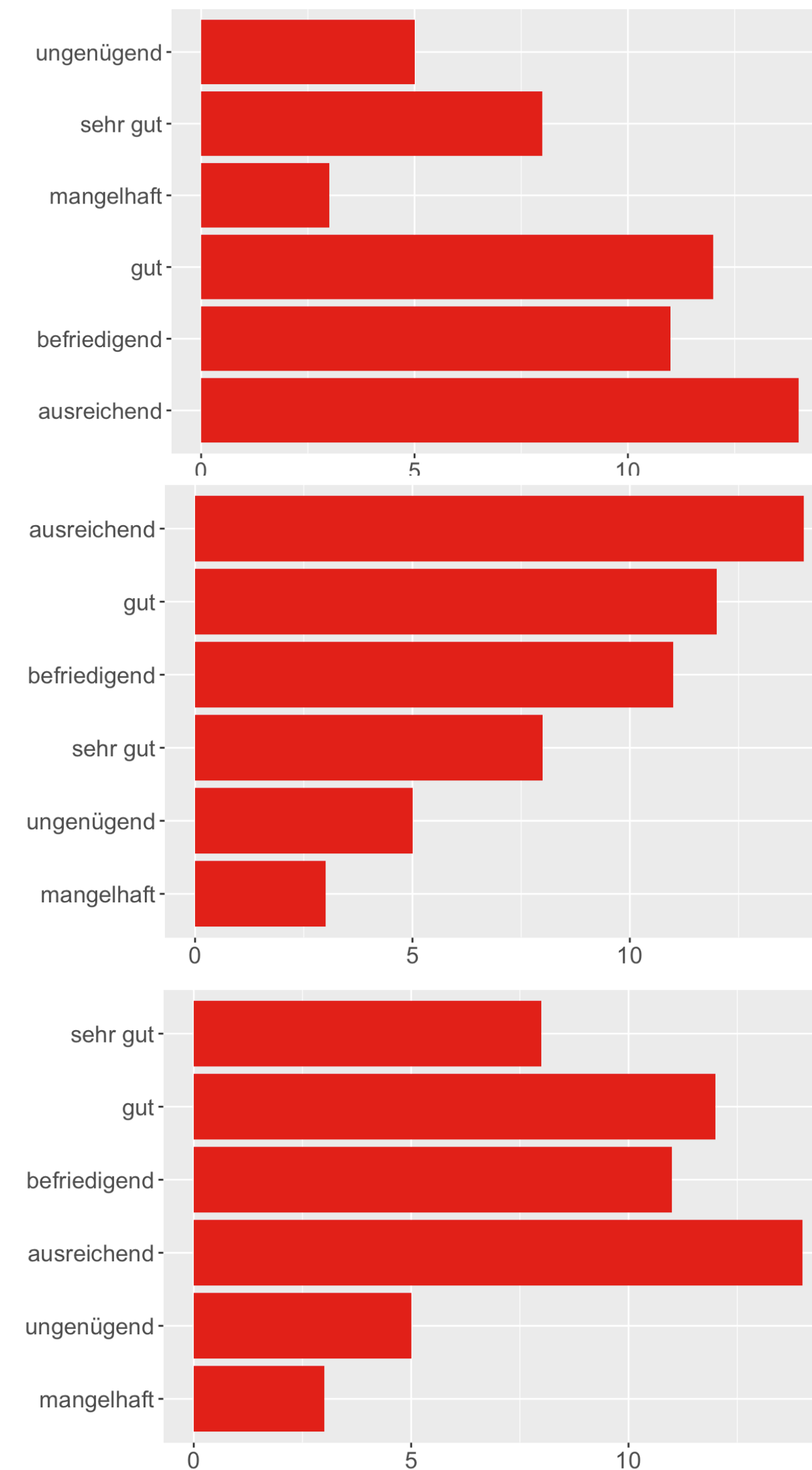
Categorical data - barplot

- Consider ordering the data by counts (no natural order of categories for nominal data)
- Beware of selecting the proper scales for plotting!



Categorical data - barplot

- Random order
- Order by increasing / decreasing counts
- Natural order of ordinal data



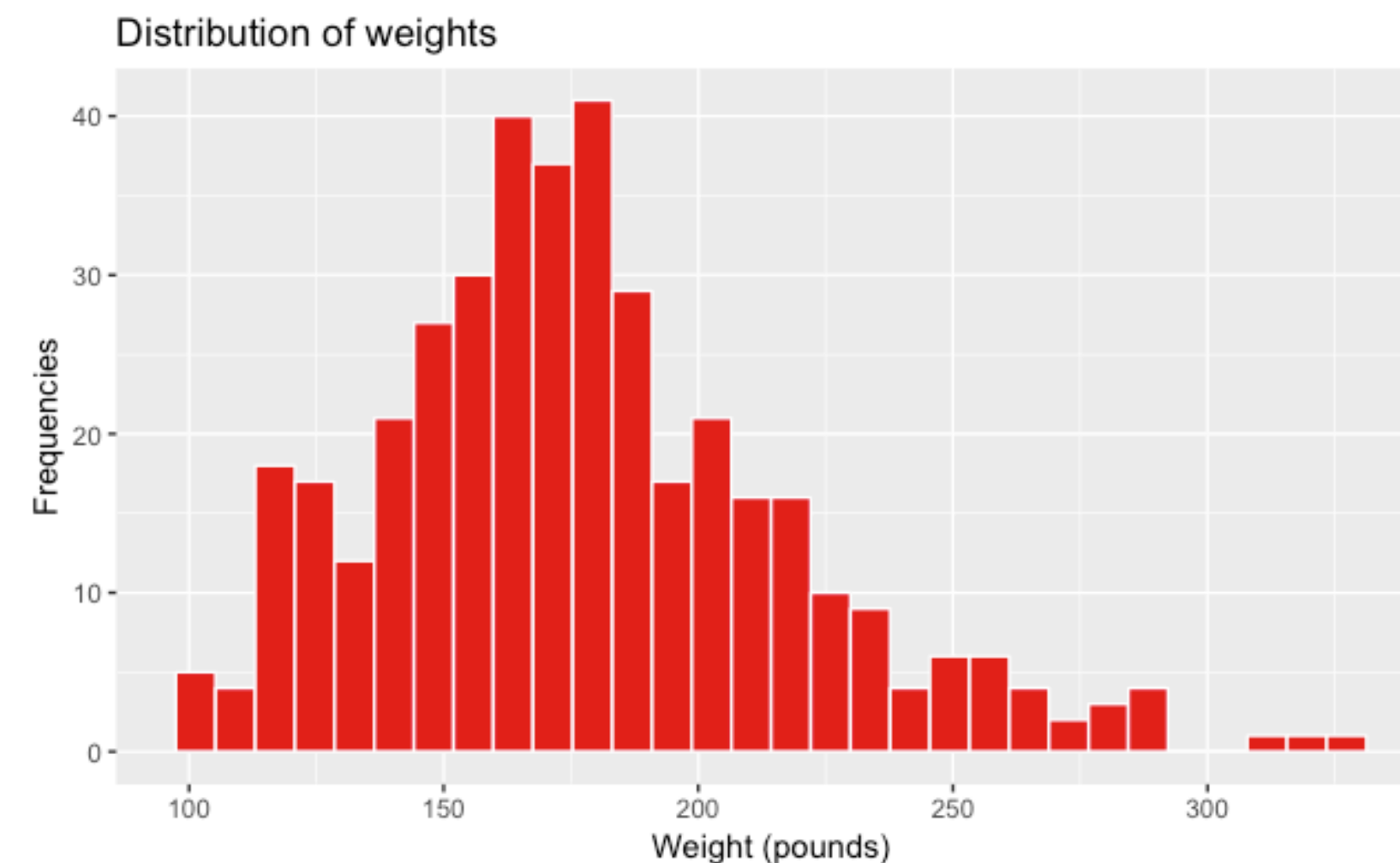
← Mode

Numerical variables

- Numerical data are instances of underlying **random variable**
- Random variables X have
 - **Density** distributions $p(X)$
 - **Expectation values** $E(X)$
 - **Variances** $Var(X)$

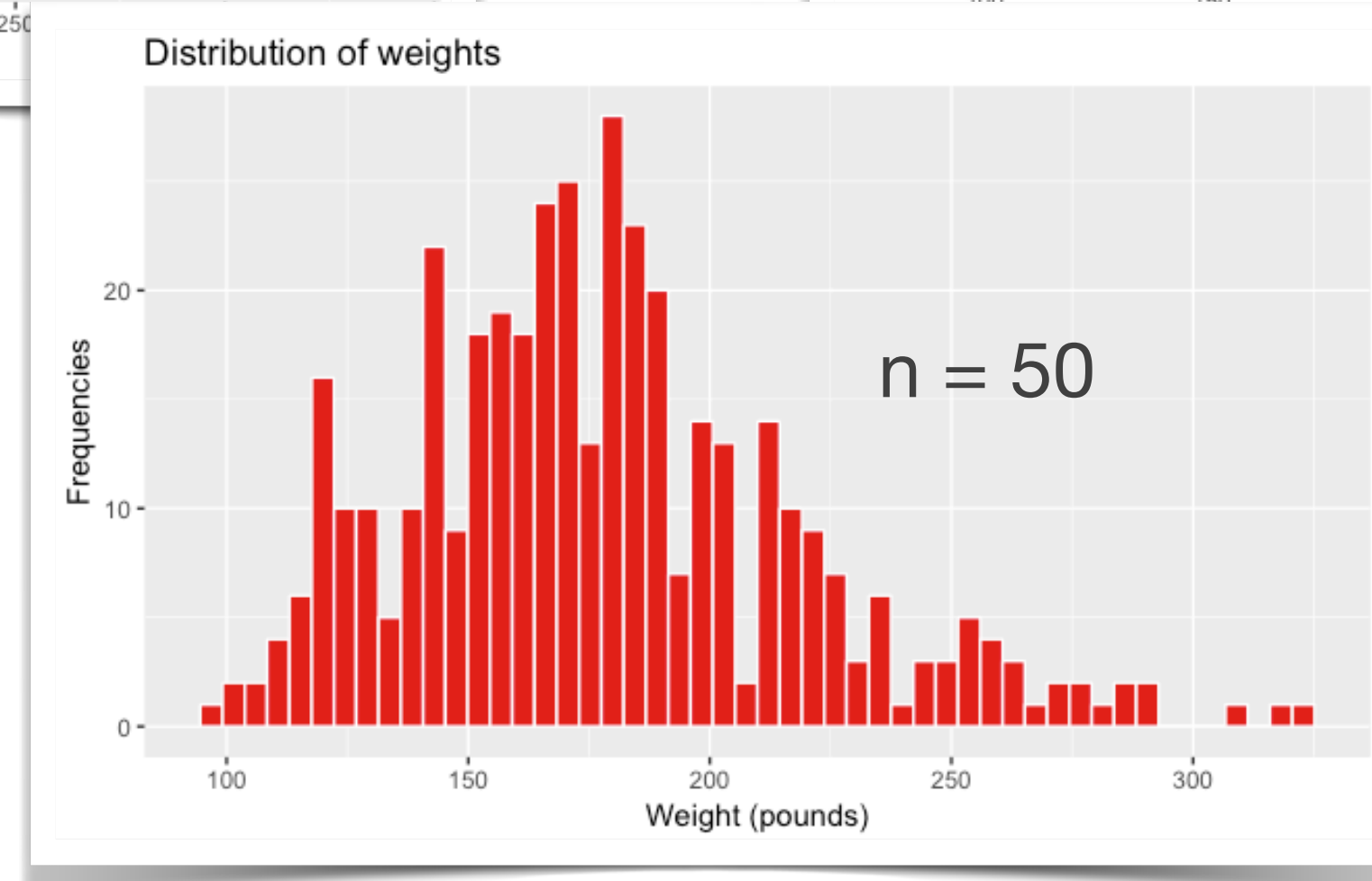
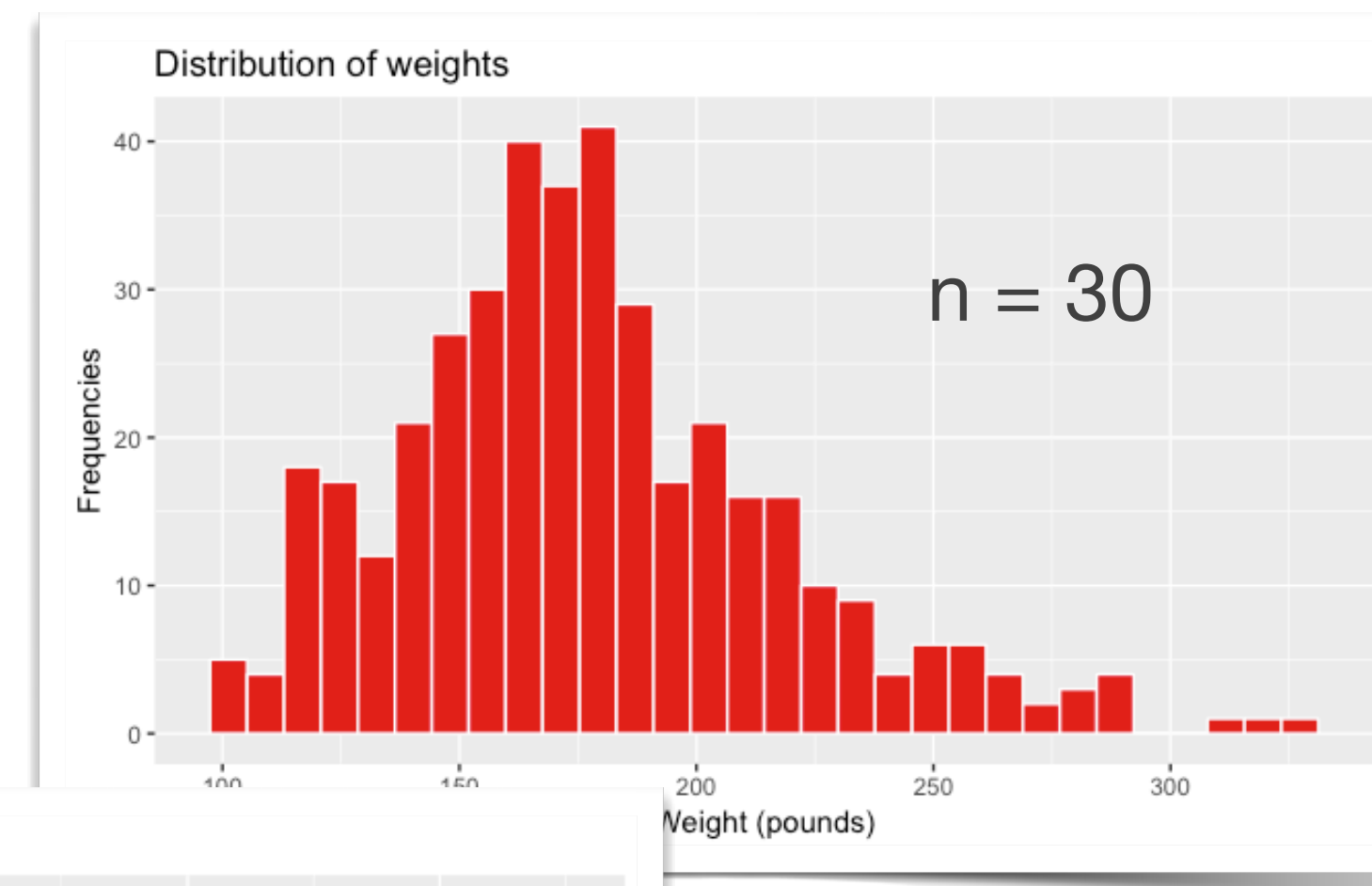
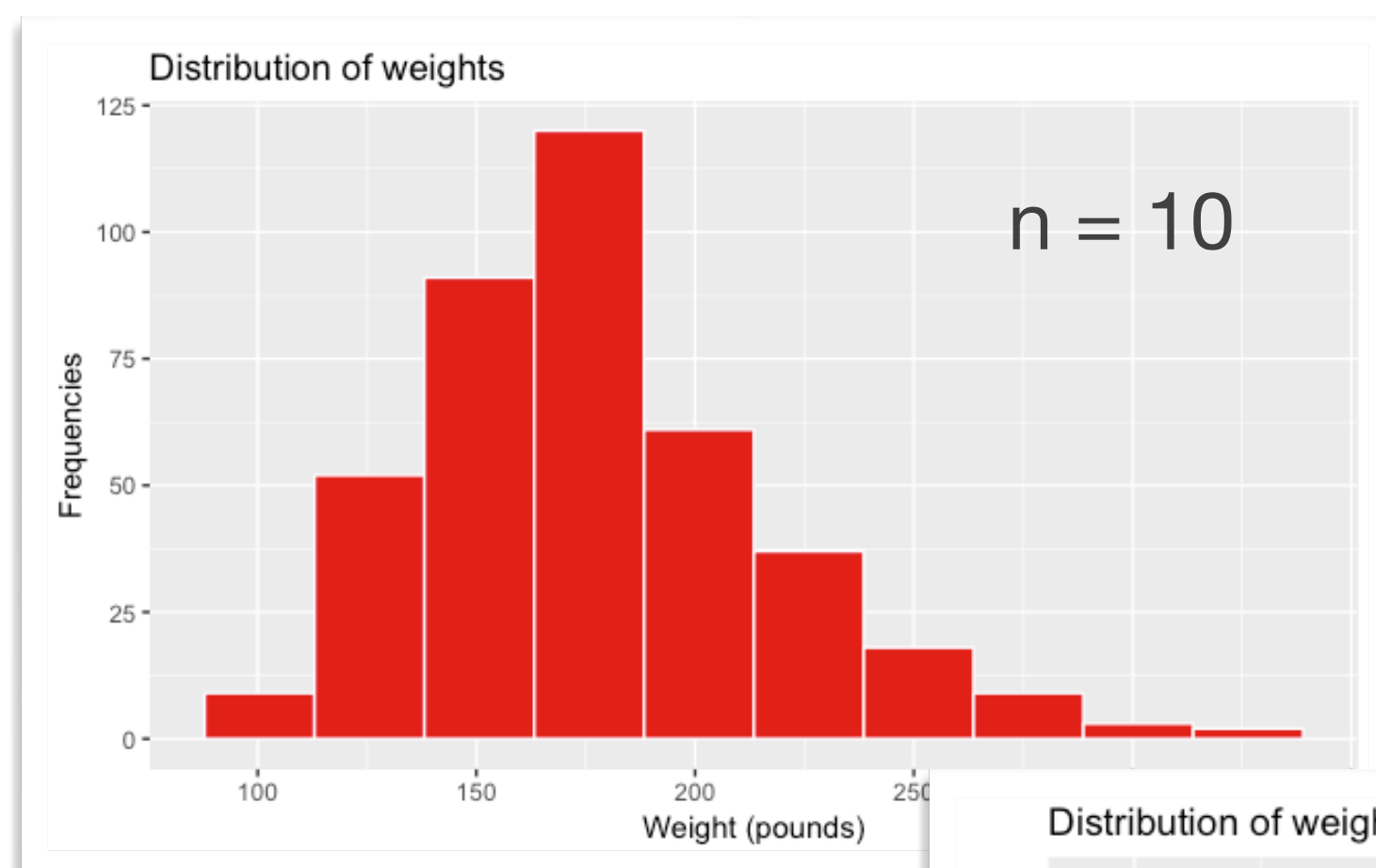
Numerical data - histograms

- Categorical data → counts (**barplot**)
- Numerical data → counts within intervals (**histogram**)
 - define **discrete intervals** for numerical variable → **ordinal variable**
e.g. $[0,10)$, $[10,20)$, $[20,30)$, ...
 - count occurrences within intervals and plot
- histograms represent the **distribution of the variable**



Numerical data - histograms

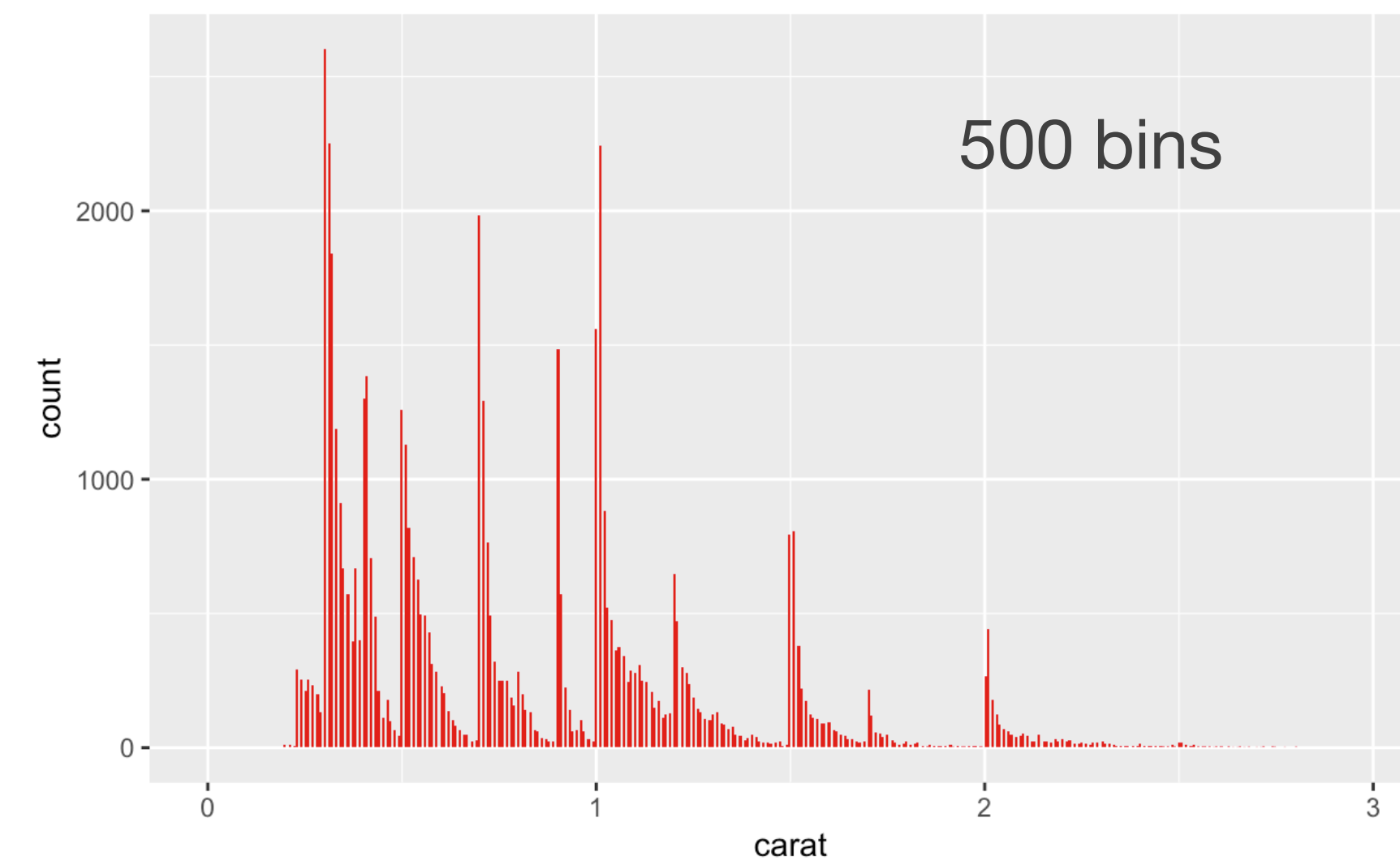
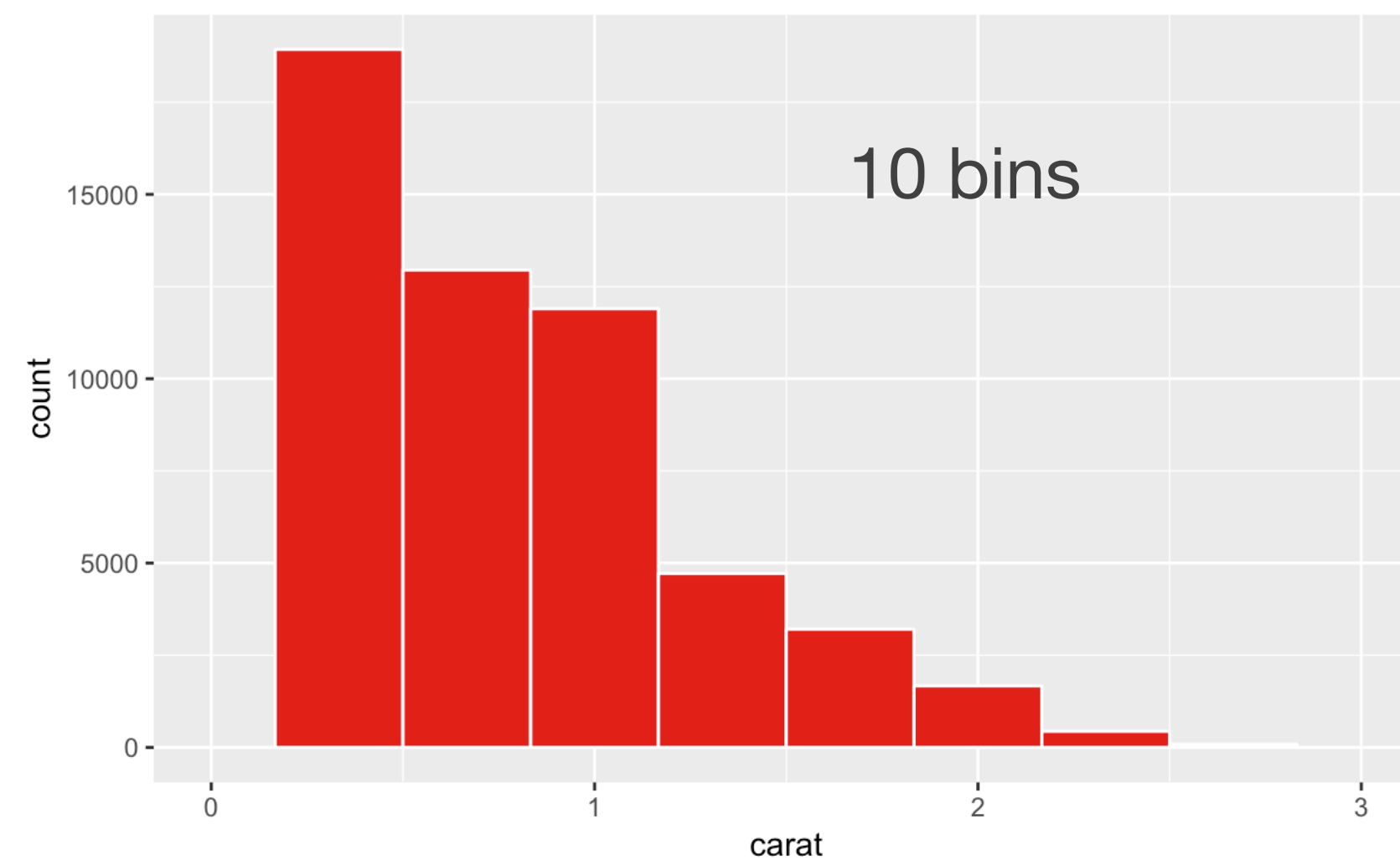
- Right choice of interval depends on the data type
- **Number of bins has a strong impact on appearance of plot!**



Numerical data - histograms

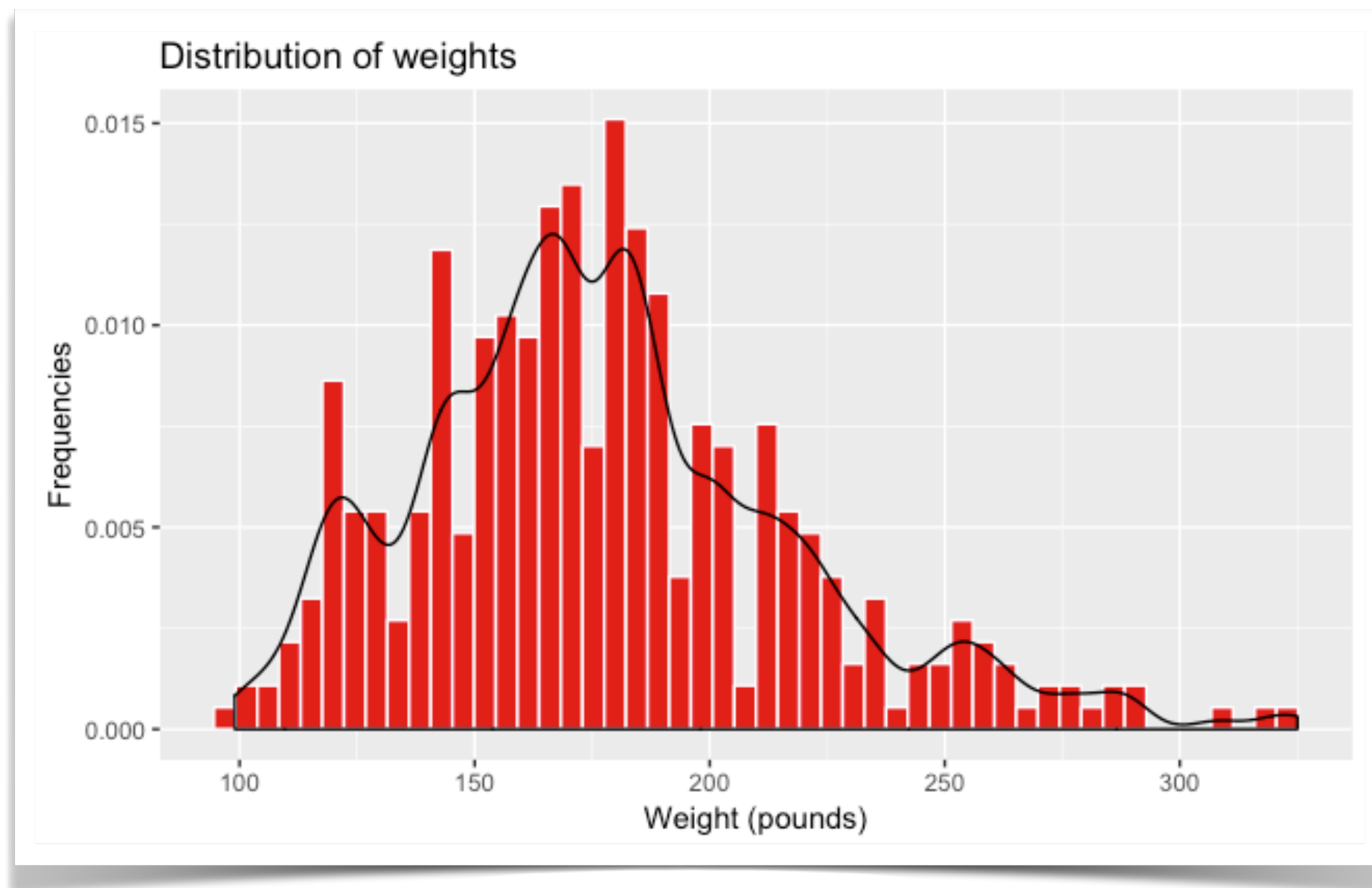
- pattern becomes visible at high resolution
- peaks around integer values (why?)
- tail on the right of integer values (why?)

Distribution of carat values for diamonds

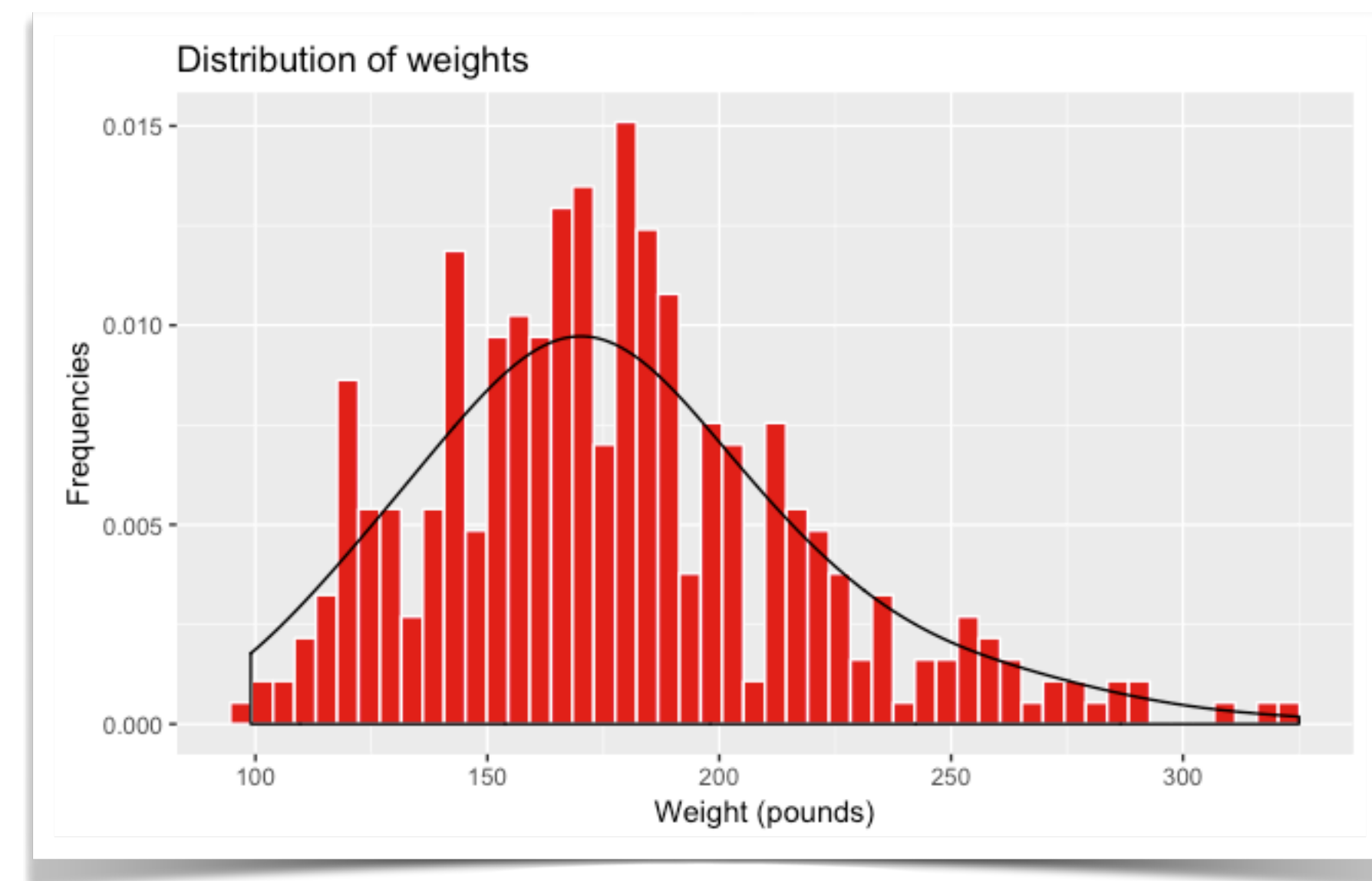


Numerical data - histograms

- Frequency distributions (= histograms) can be shown using a smoothed **density curve**
- Smoothing depends on the bandwidth (~size of the interval over which to smooth)



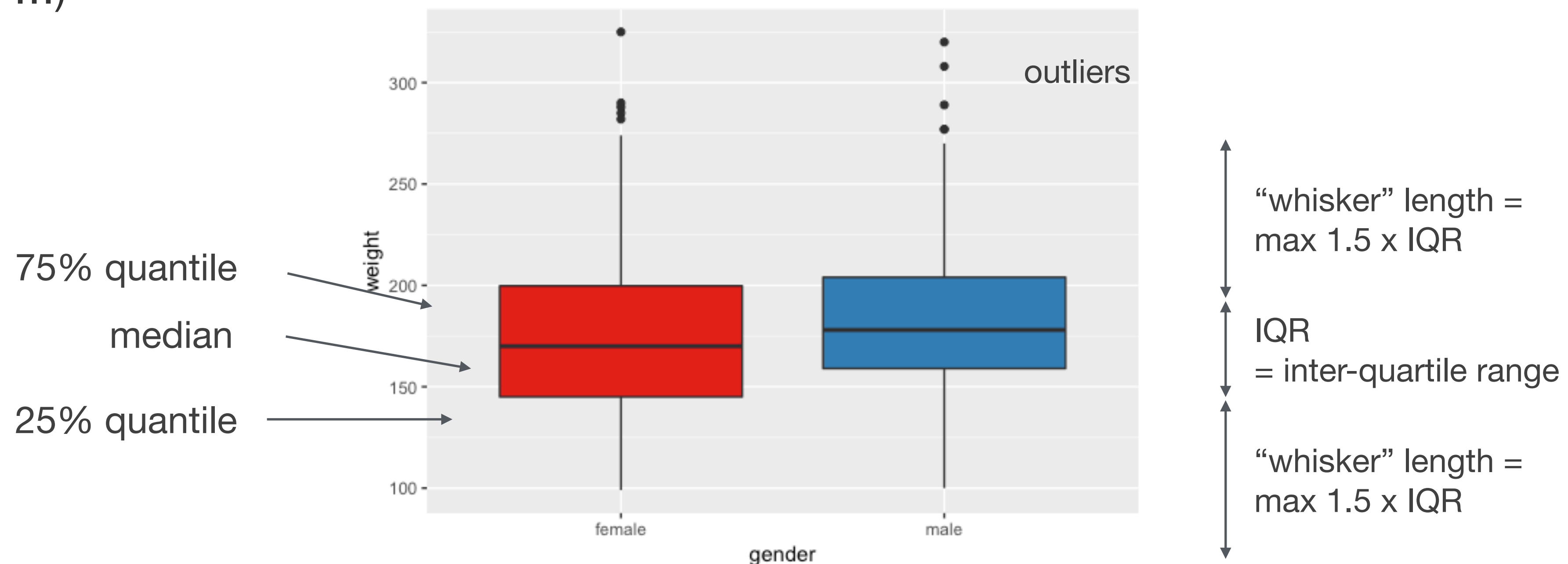
bandwidth = 5



bandwidth = 20

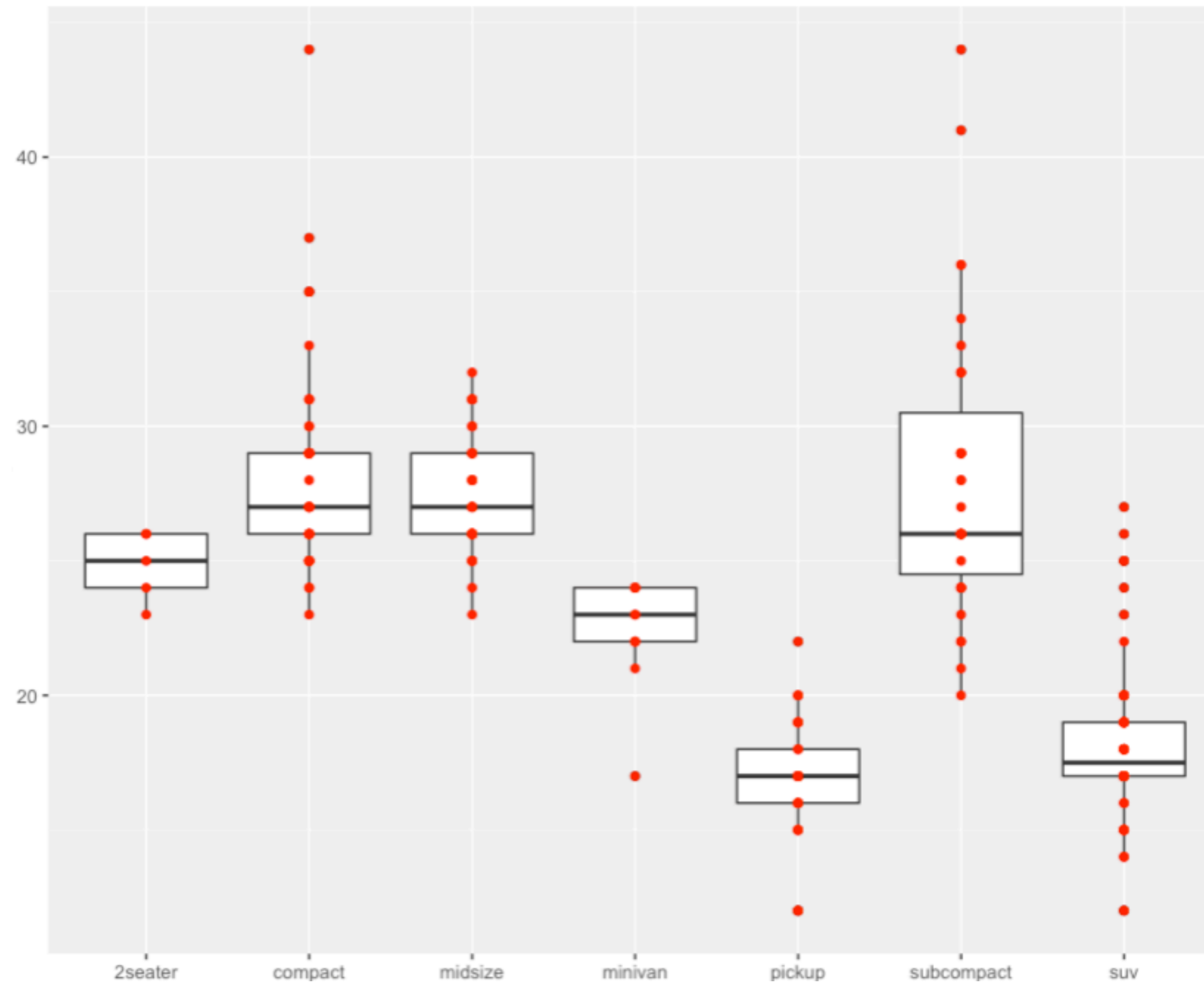
Numerical values - boxplots

- Boxplot give an indication on the shape of the distribution (median / symmetry / outlier / ...)



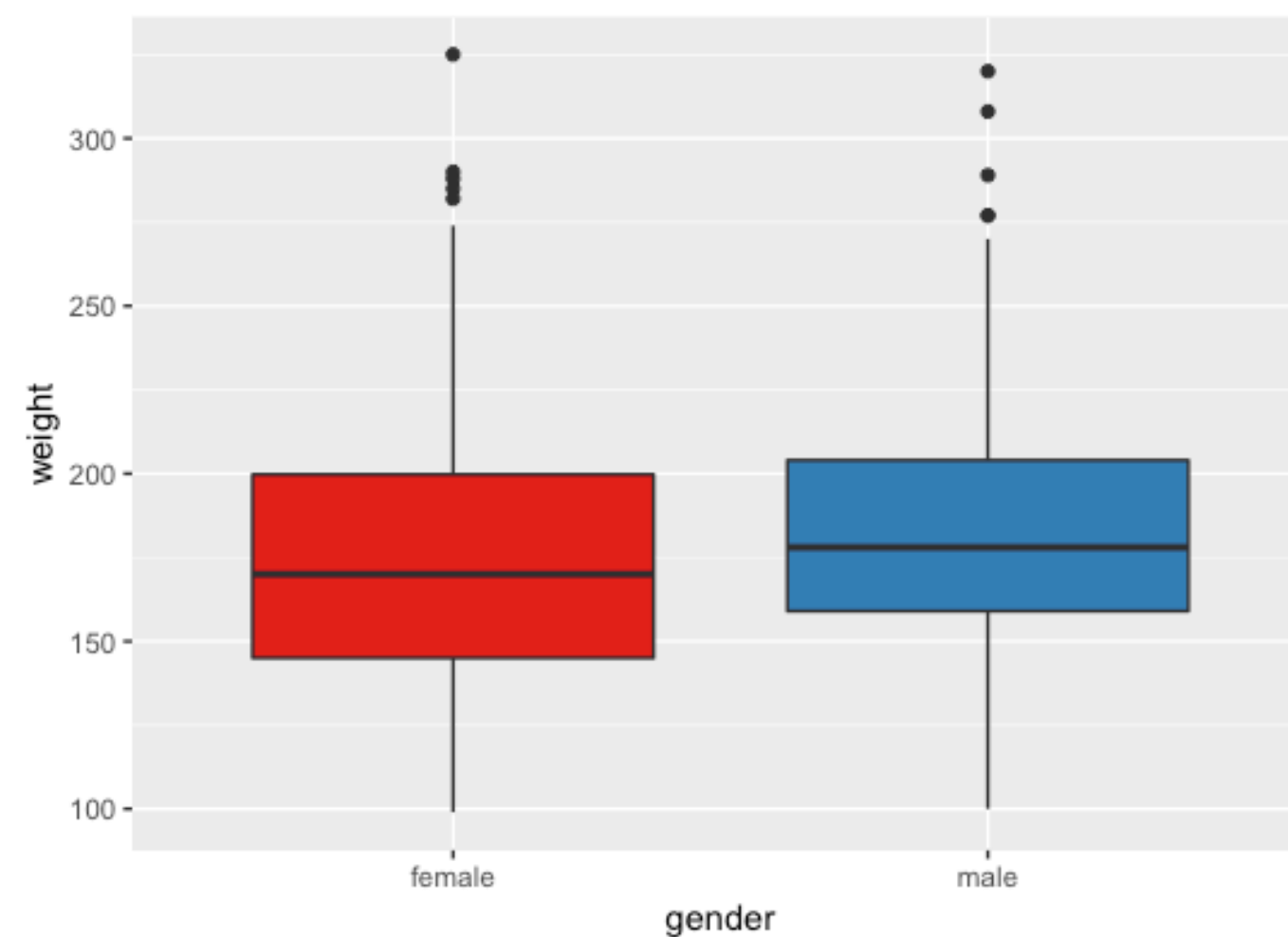
- Upper Whisker extend to the last point that is not larger than $Q75 + 1.5 \times IQR$
- Lower Whisker extends to the last point that is not smaller than $Q25 - 1.5 \times IQR$
- Whisker does not go beyond maximum or minimum value ! (Hence both whisker can have different length $< 1.5 \times IQR$)

Numerical values - boxplots

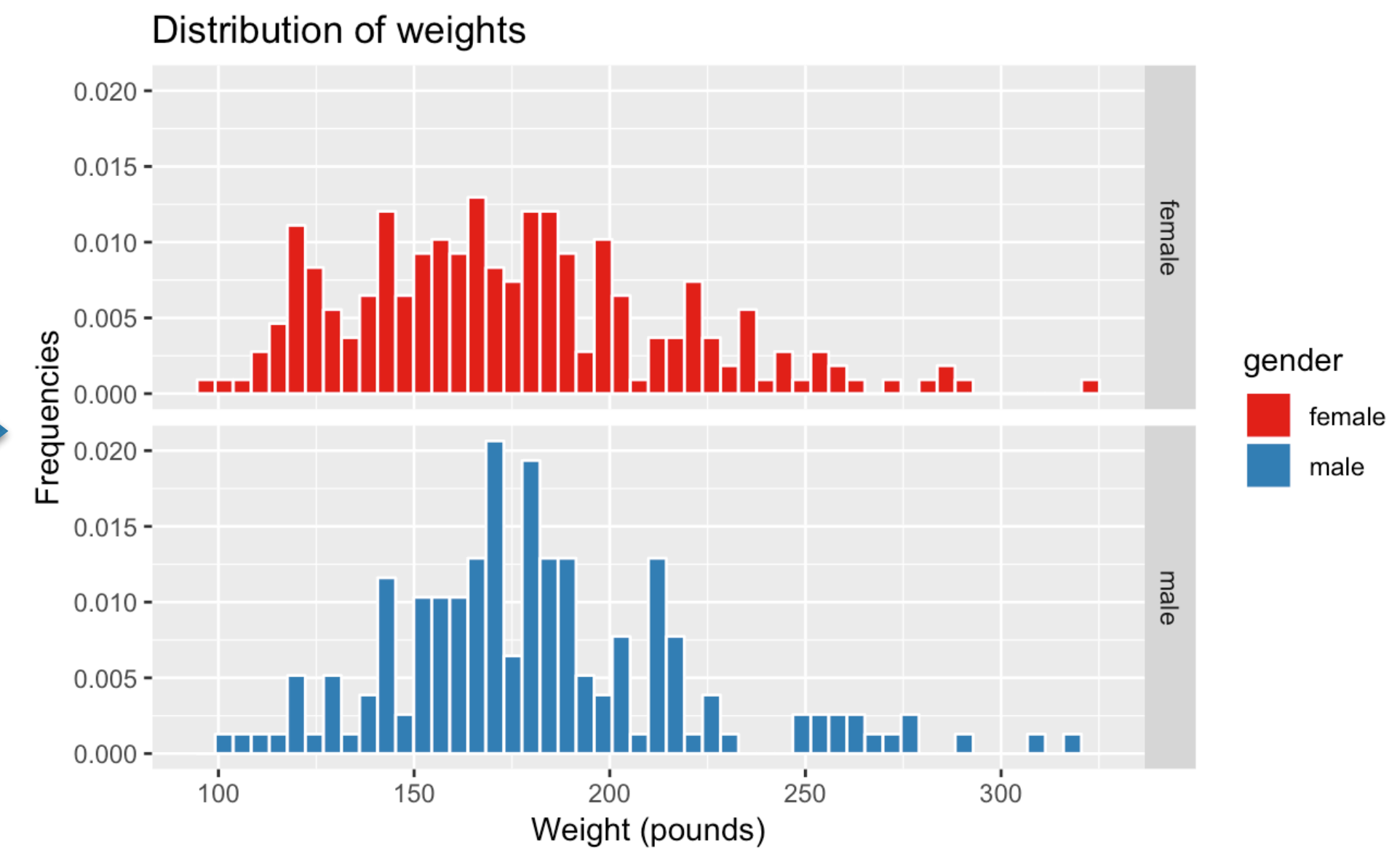


Numerical values - boxplots

Boxplot

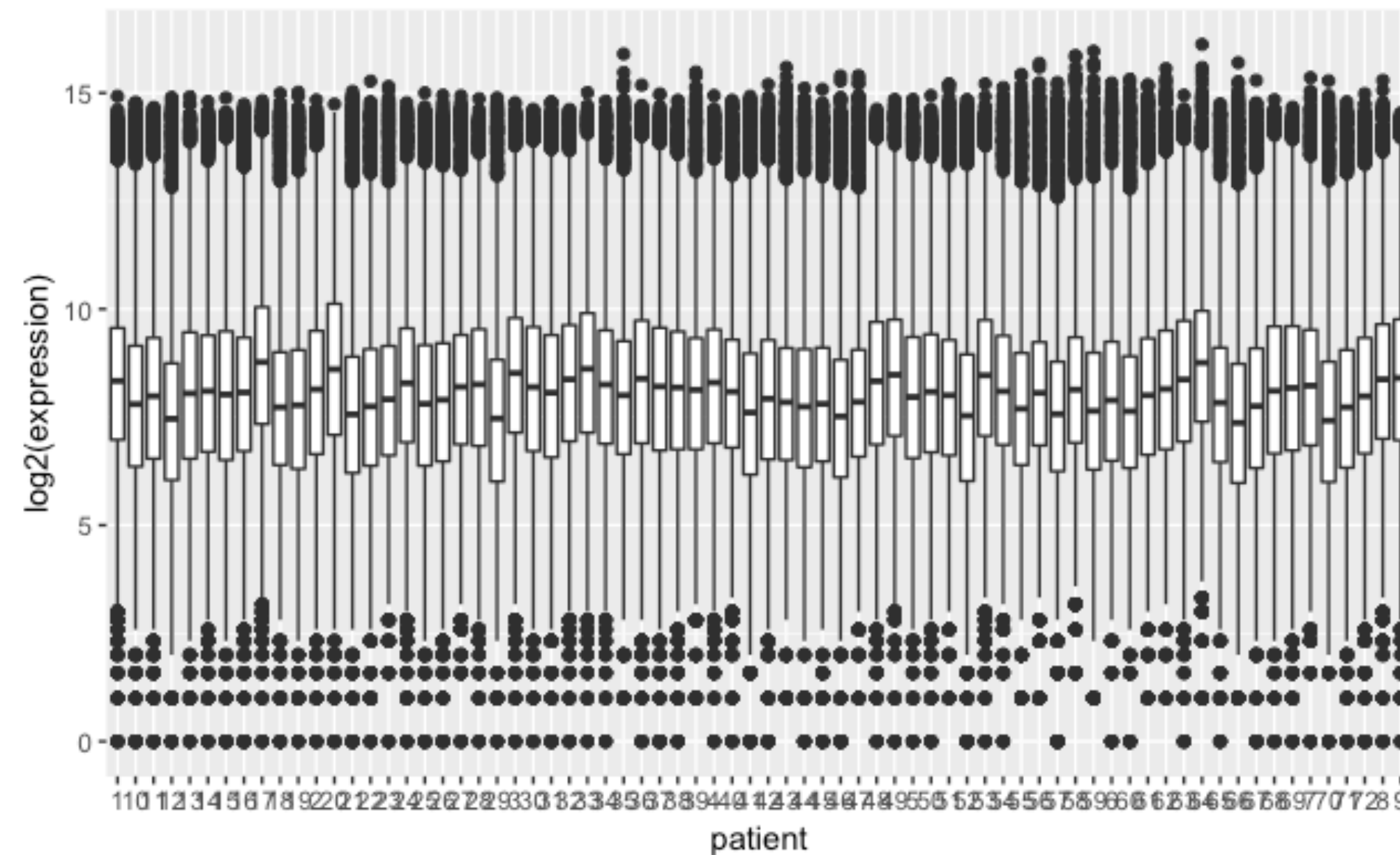
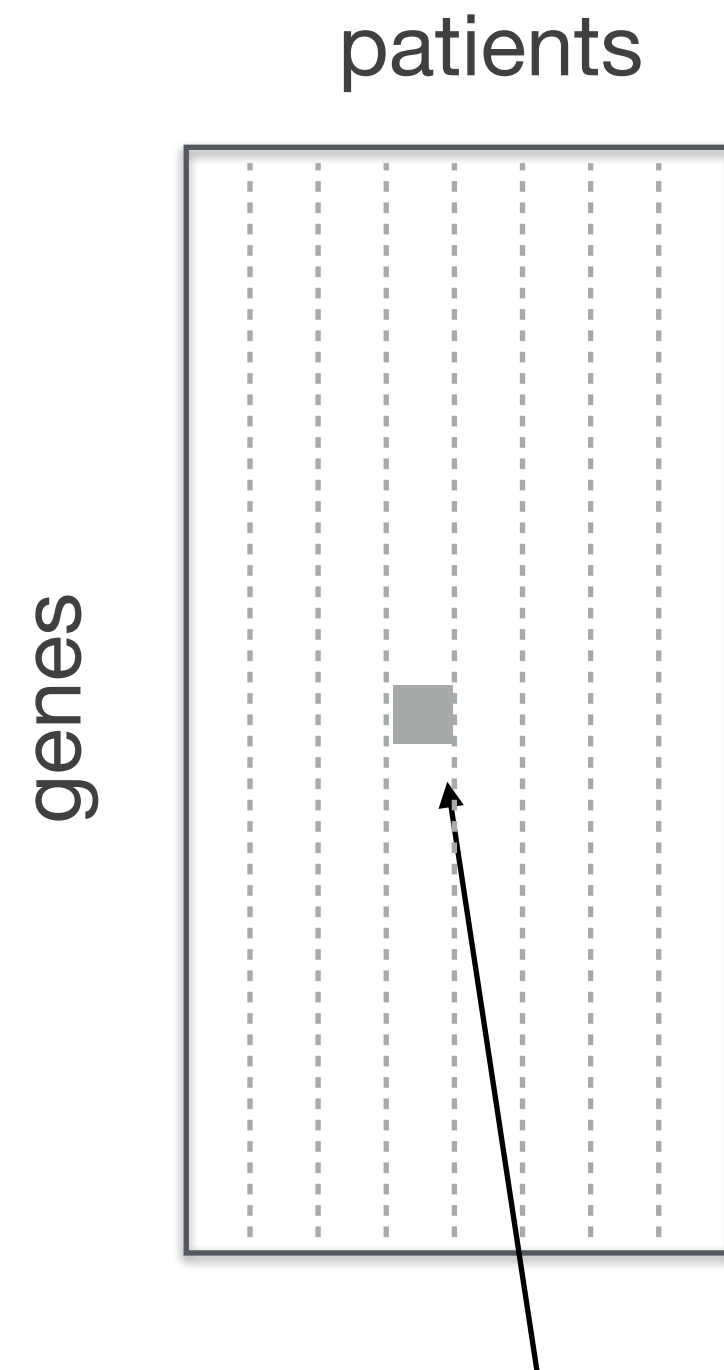


Histogram



Boxplots summarize the properties of the distribution
Usefull to compare many distributions side-by-side

Numerical values - boxplots

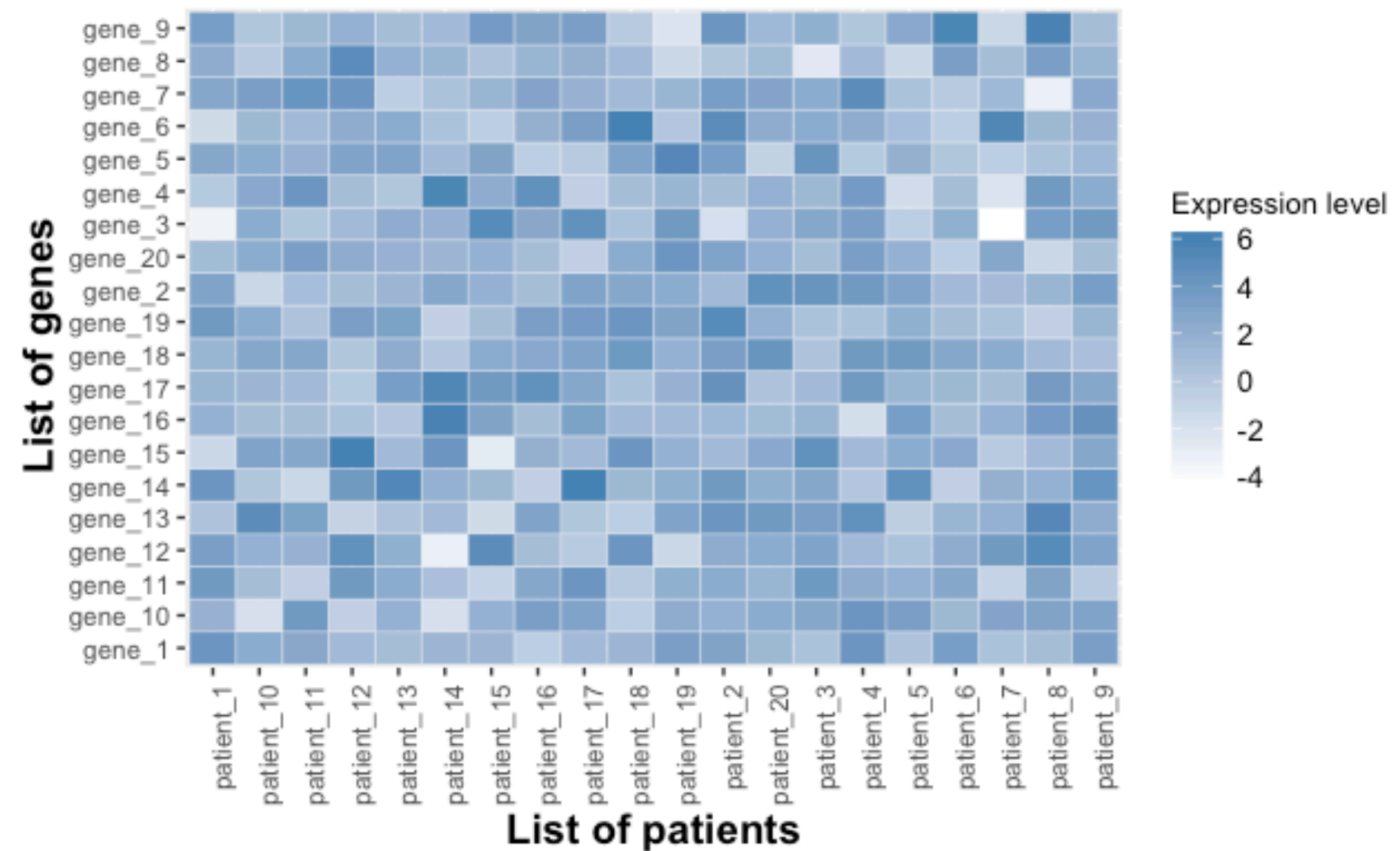
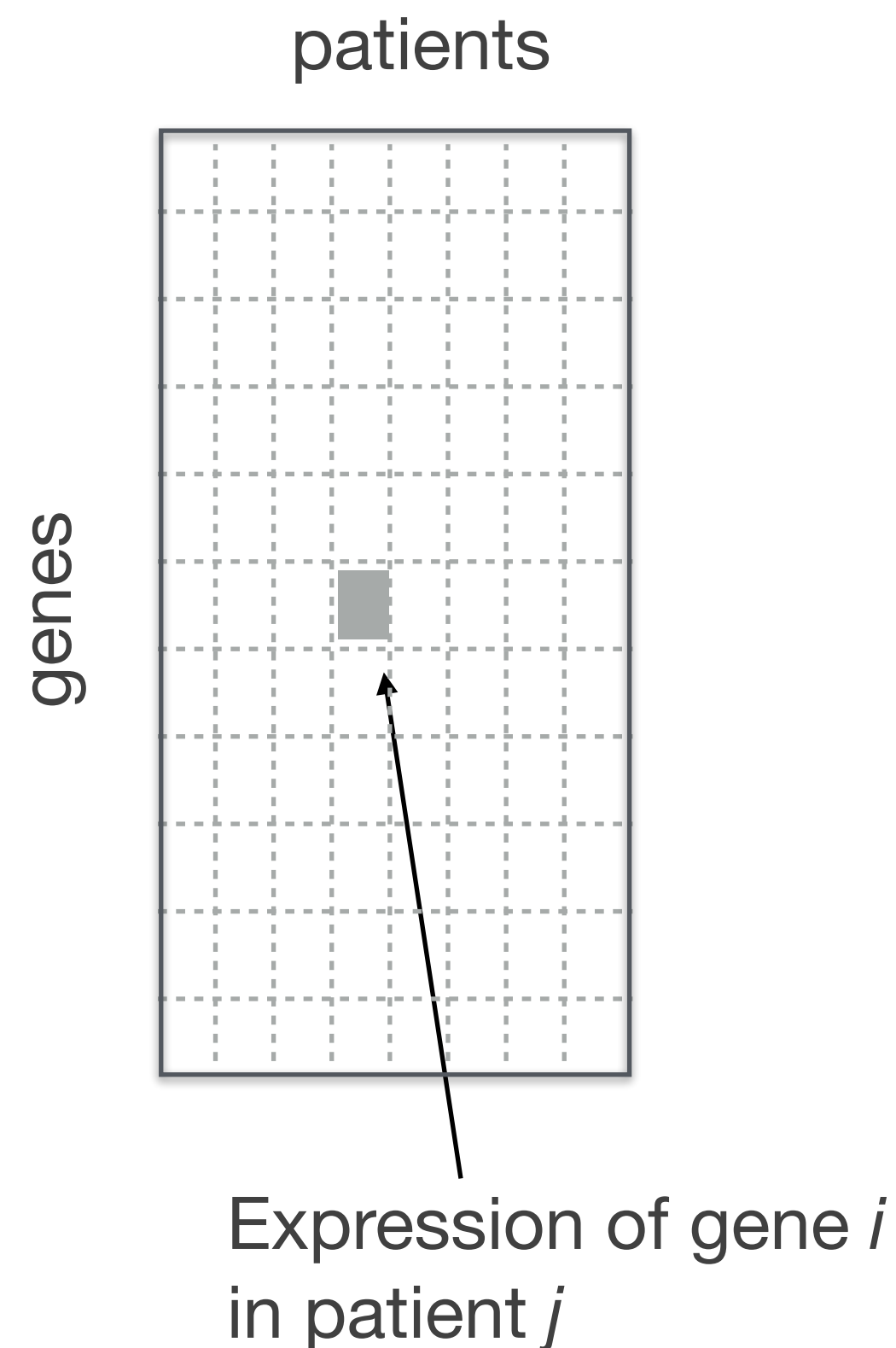


Question: do some patients have
a different median gene expression?

→ *values for individual genes are lost in this type of plot!*

Numerical Data - heatmaps

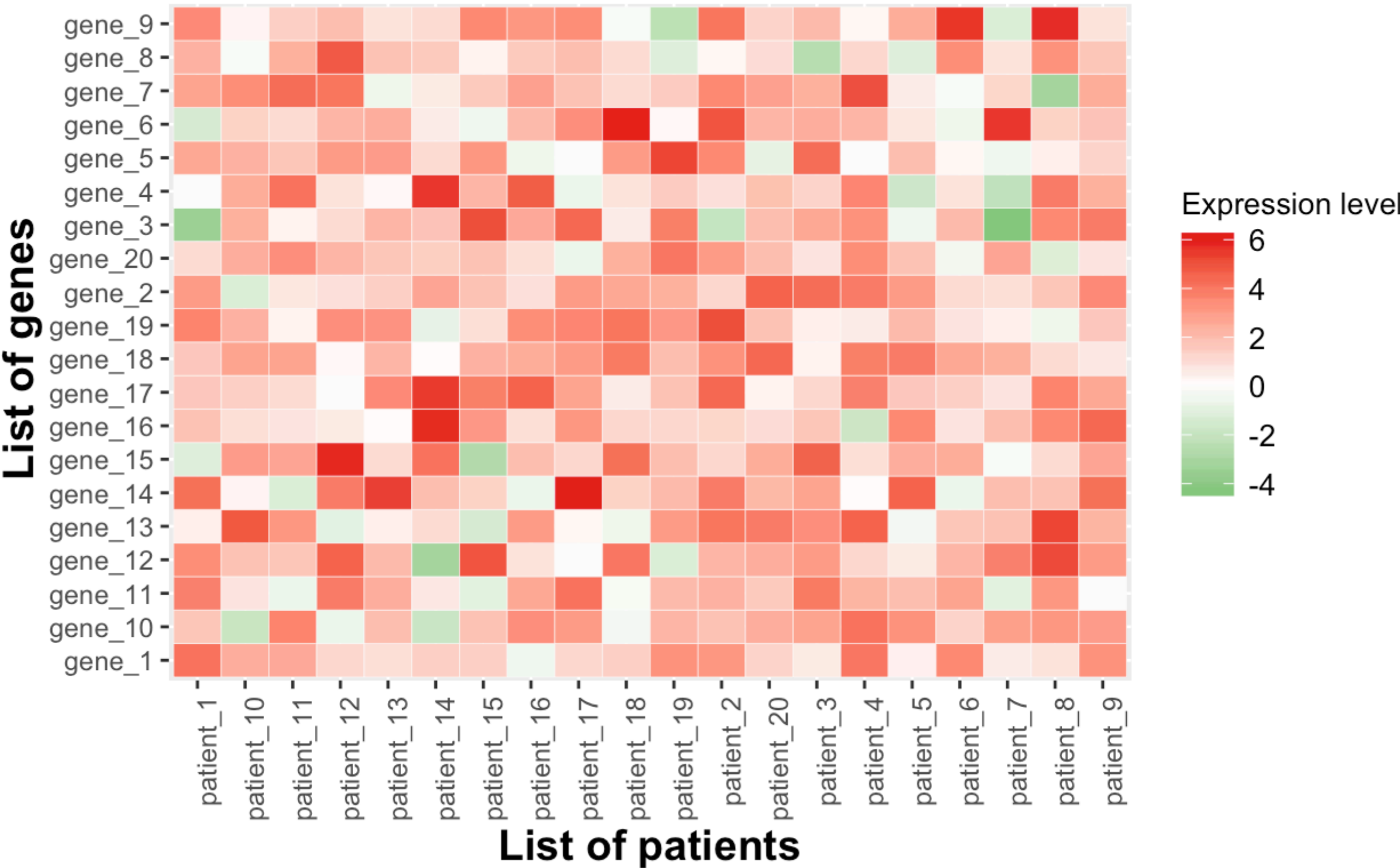
Heatmaps display numerical values in a data matrix using a color scheme



Question: do some genes in some patients
have a different gene expression?

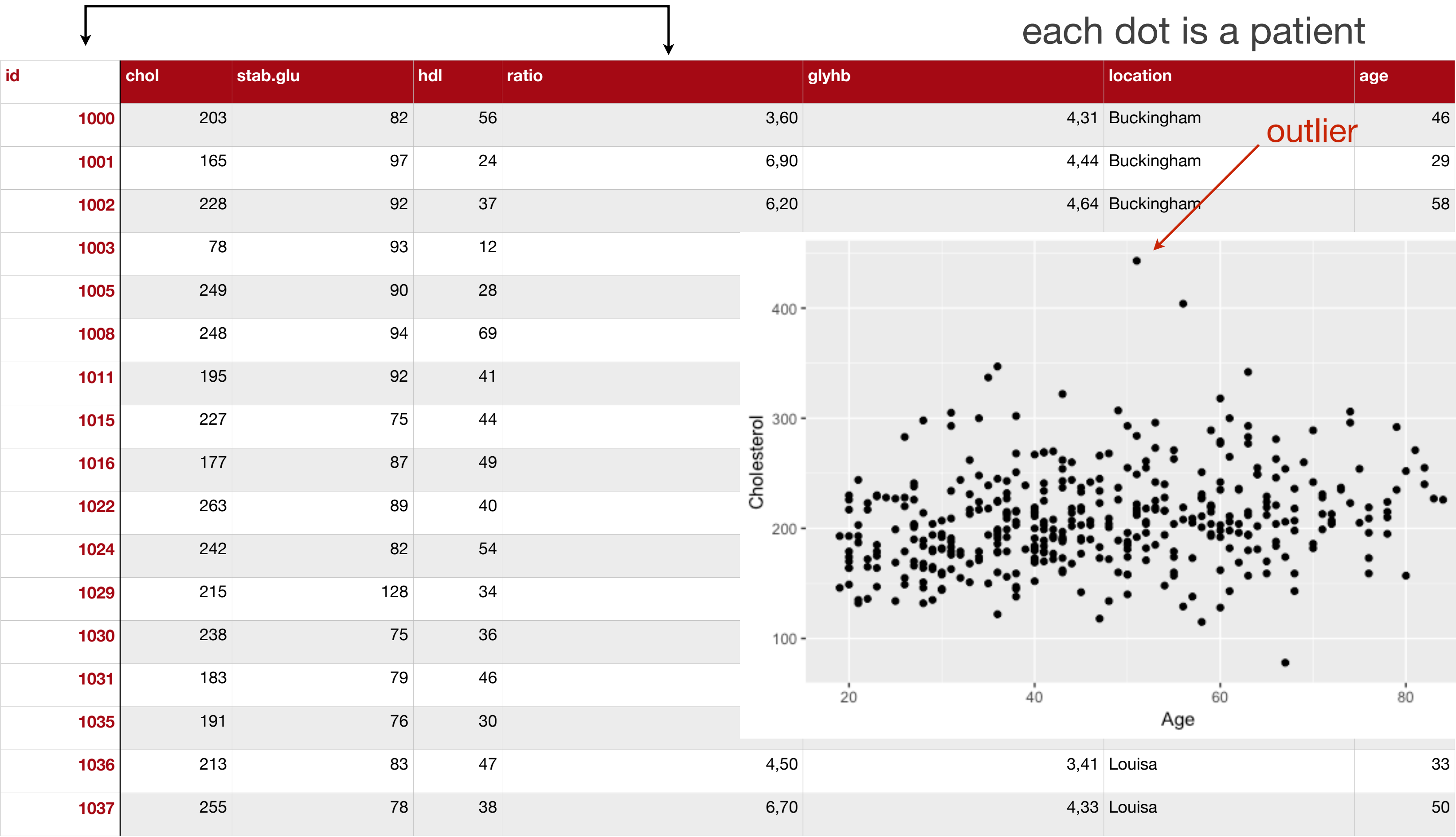
Numerical Data - heatmaps

use symmetrical color scales for symmetrical ranges



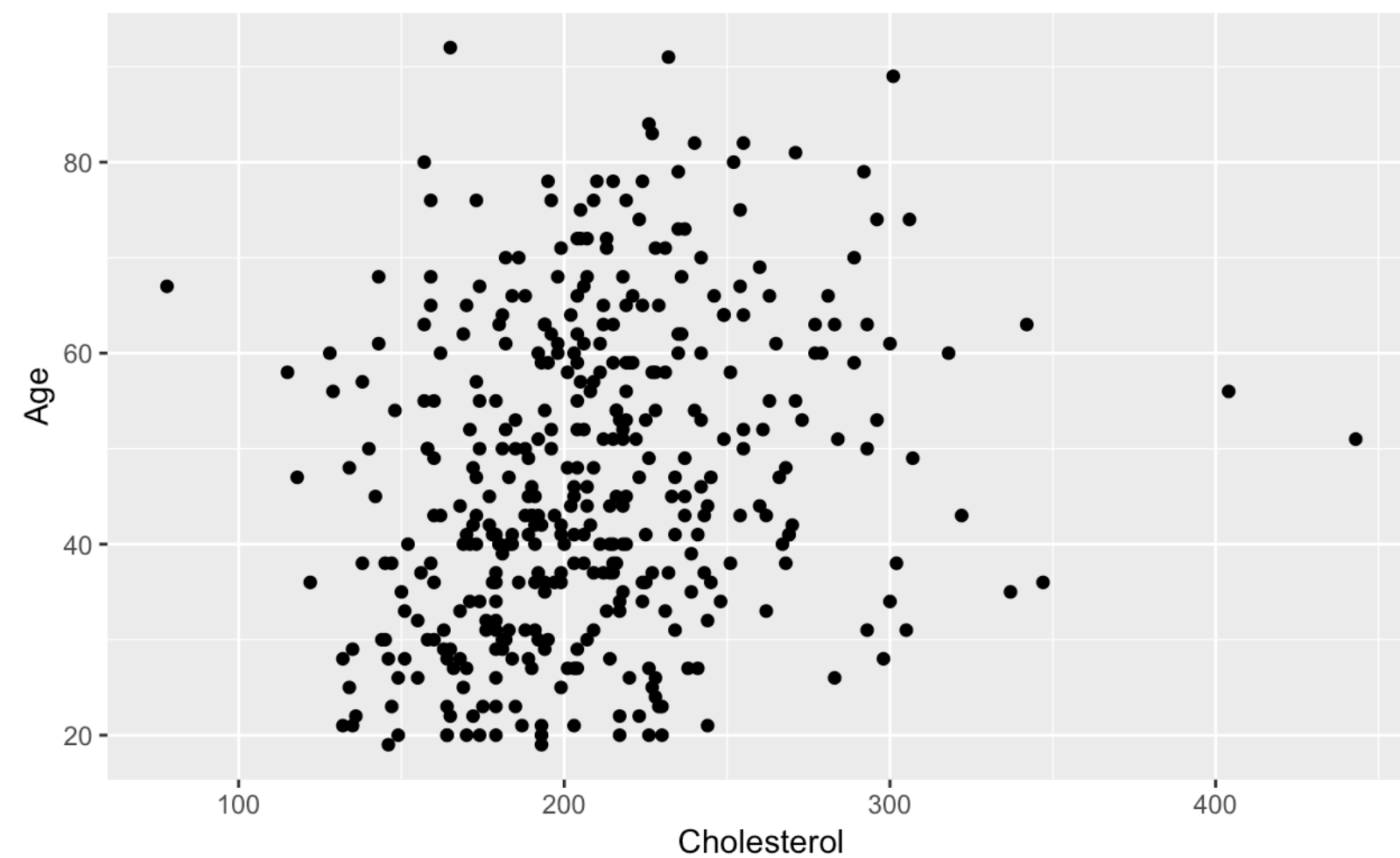
Numerical data - scatter plots

any relation between age and cholesterol?

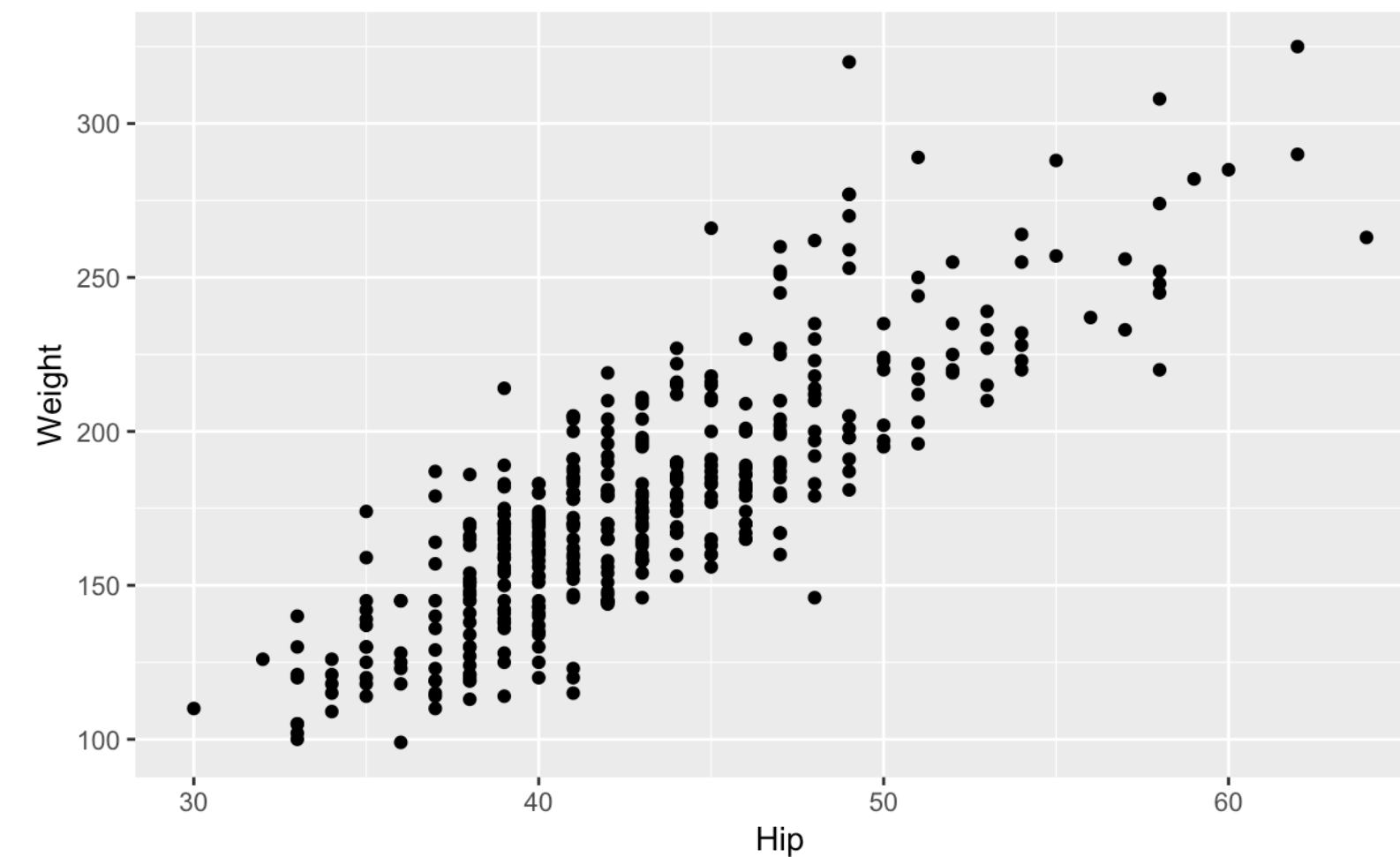


Numerical data - scatter plots

- we will **quantify** this relationship in terms of **covariance / correlation** and determine how **significant** this relationship is!



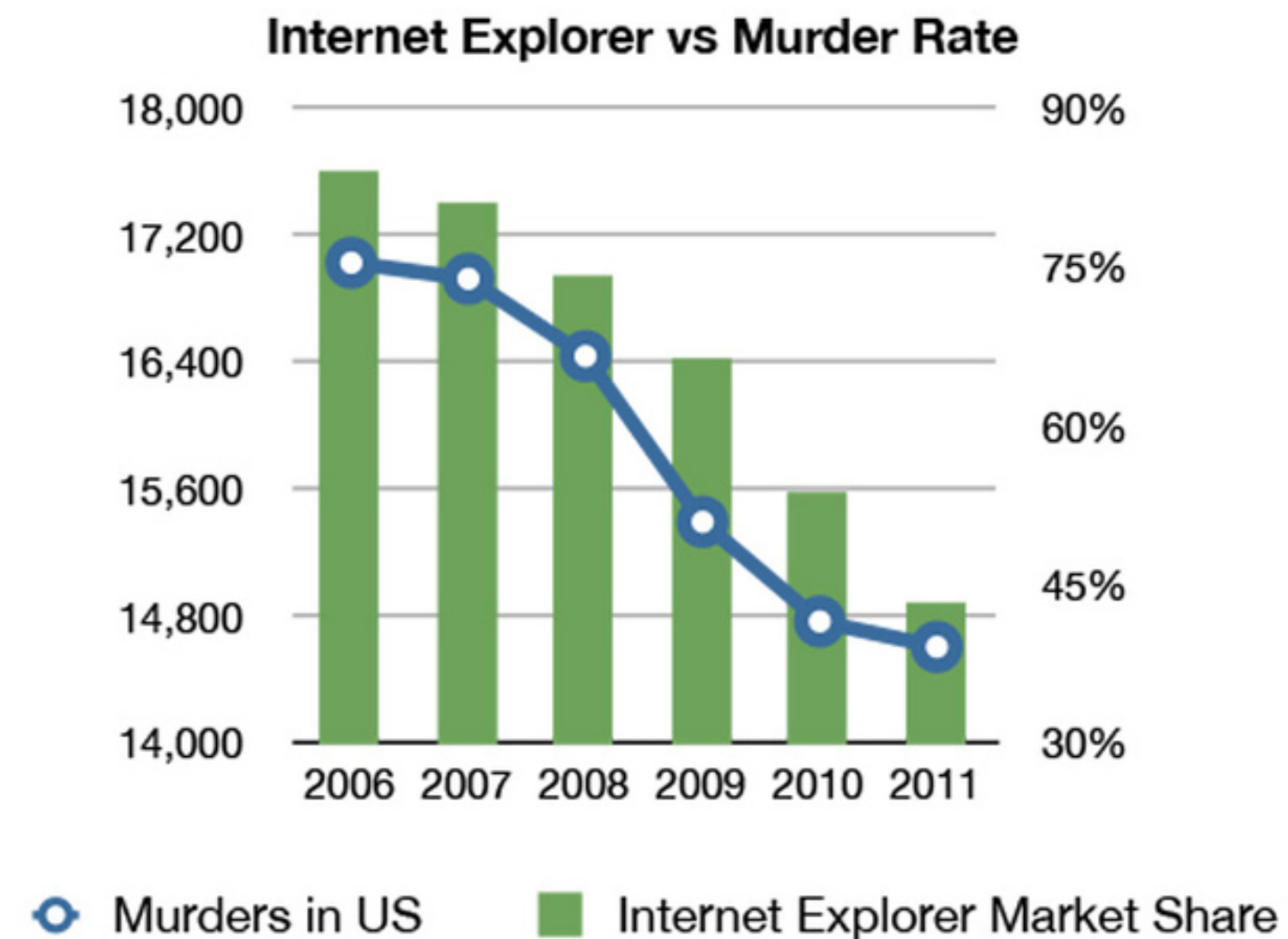
Weak relationship
between Cholesterol and age



Strong relationship
between Hip and Weight

Numerical data - scatter plots

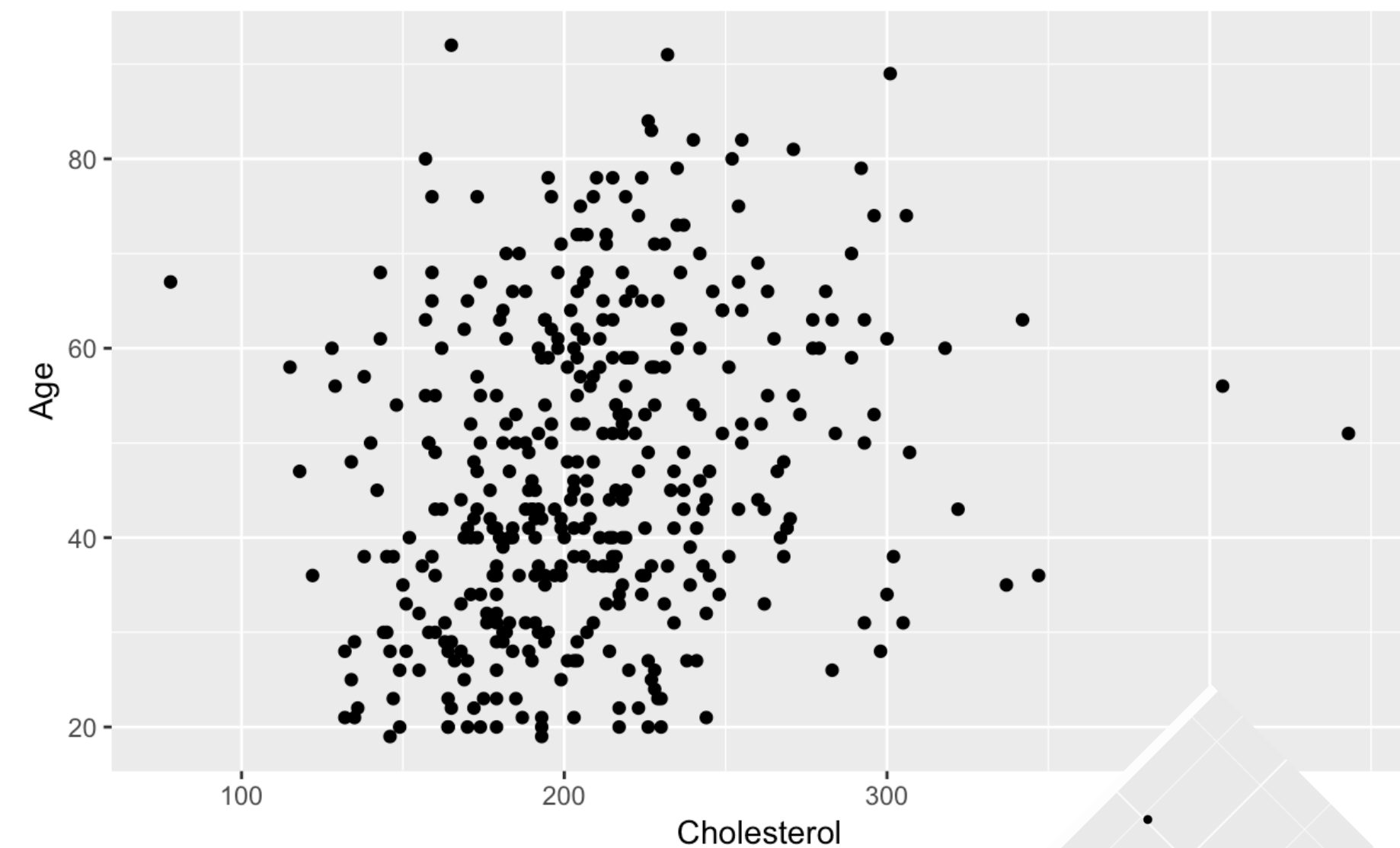
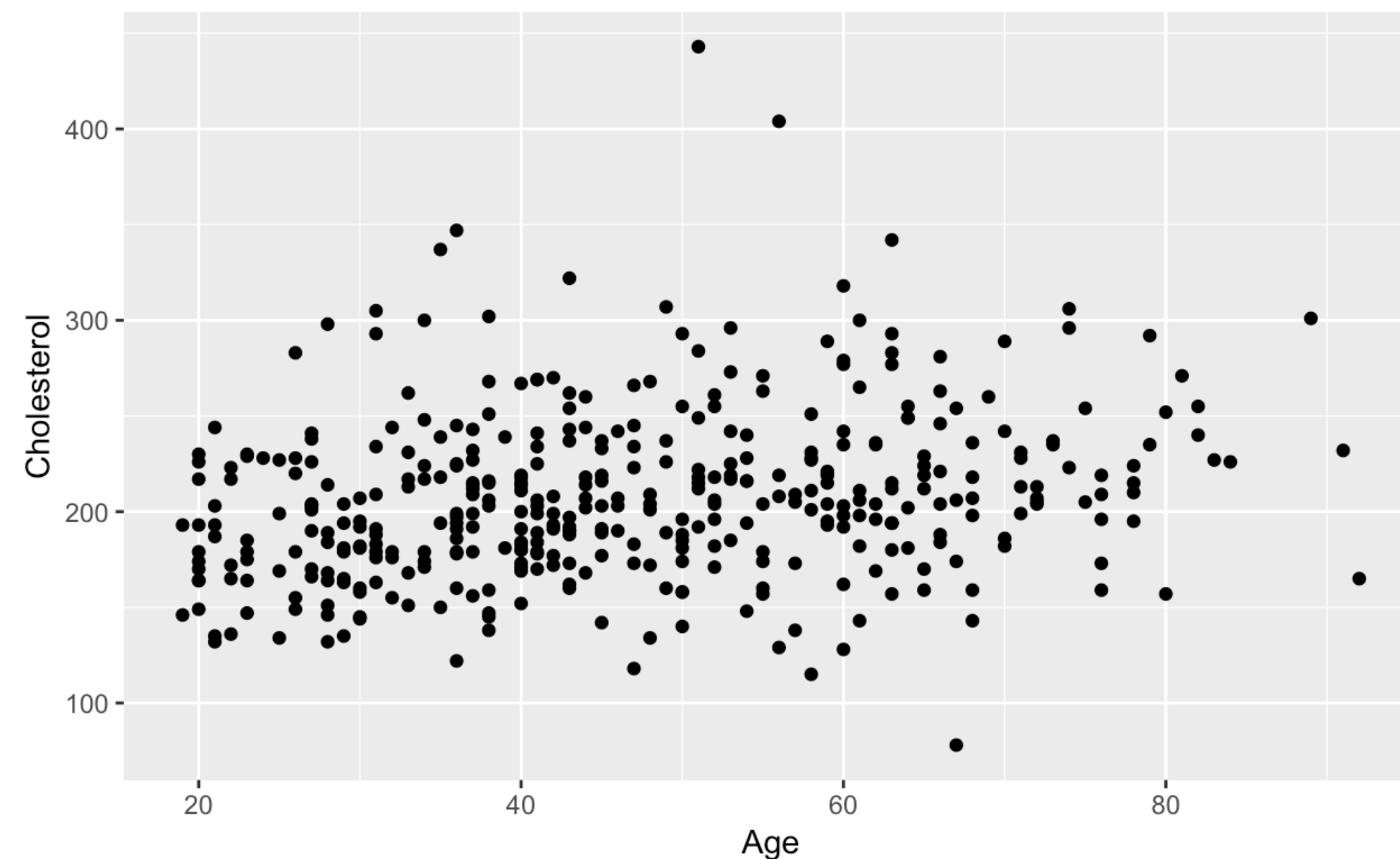
- Do not over-interpret scatter plots!
- Existence of relation between variables does not mean that there is a causal relationship between them!
- Correlation is NOT causality!!



<http://www.tylervigen.com/spurious-correlations>

Numerical data - scatter plots

- In scatter plots, the x and y axis are exchangeable
- To avoid interpreting $x \rightarrow y$, **diamond plots**



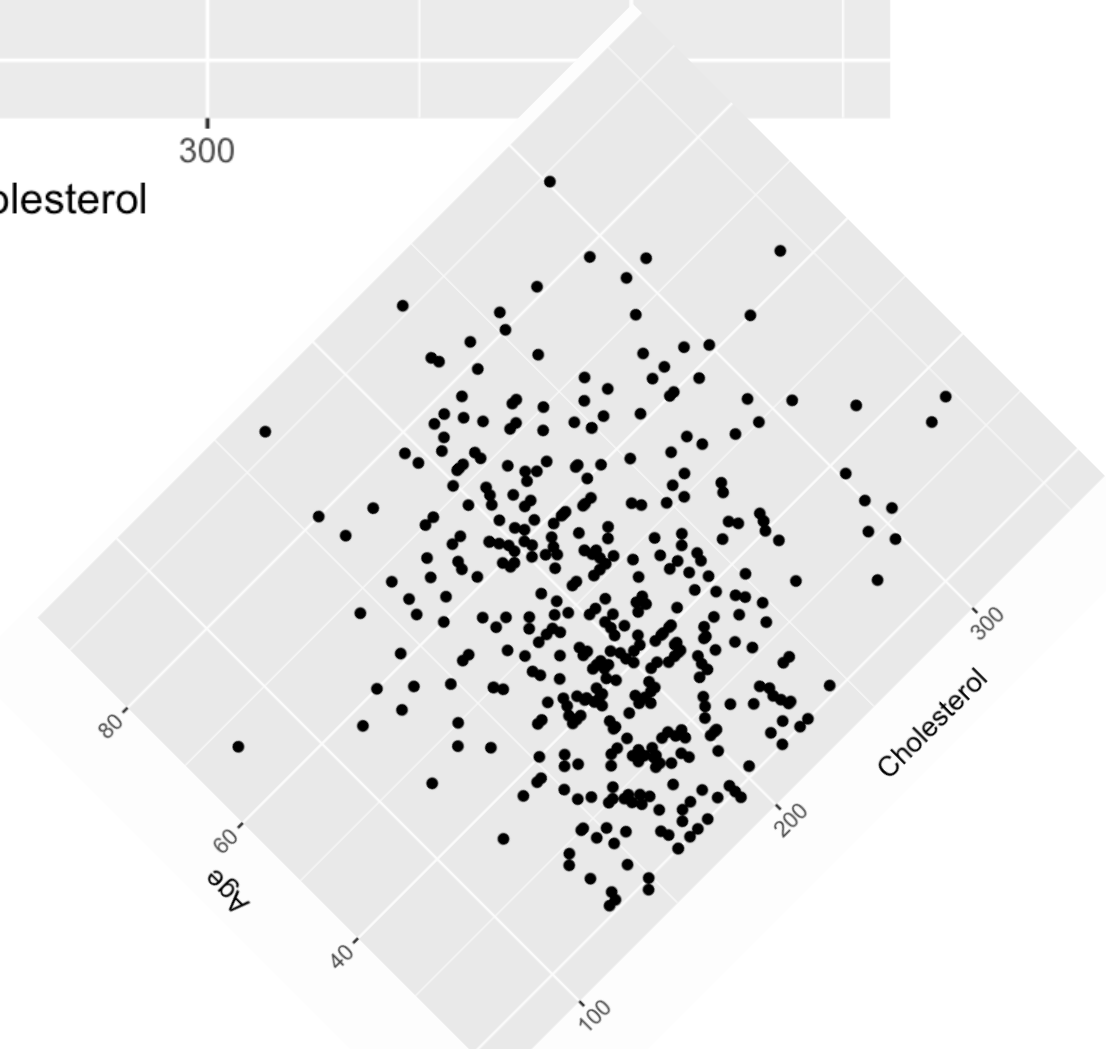
arXiv.org > cs > arXiv:1809.09328

Computer Science > Human-Computer Interaction

Why scatter plots suggest causality, and what we can do about it

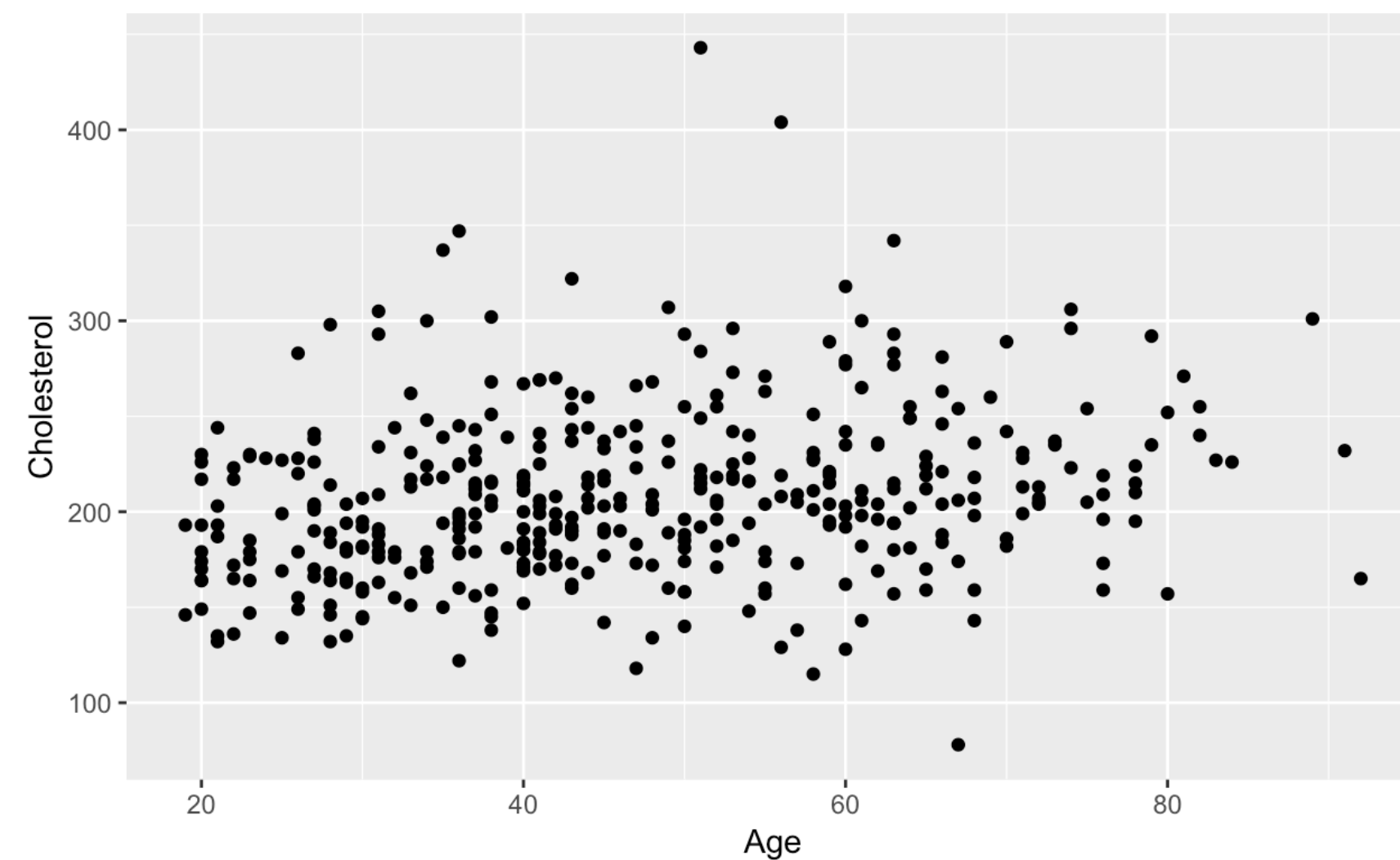
Carl T. Bergstrom, Jevin D. West

(Submitted on 25 Sep 2018)

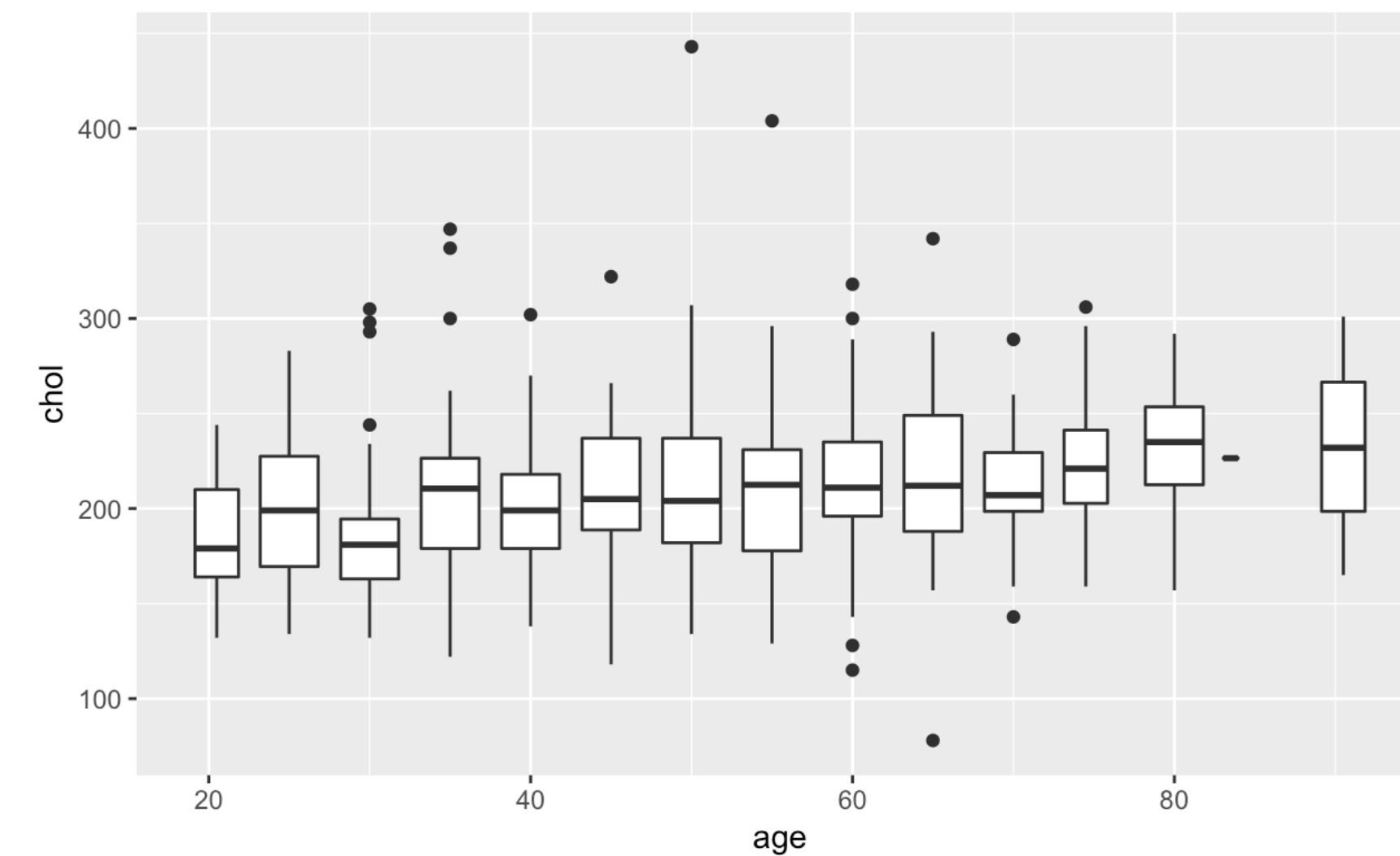


Numerical data - scatter plots

- A **continuous numerical** variable can always be transformed into an **ordinal categorical** variable through binning



Age = continuous variable



Age = ordinal variable (bins)

Numerical data - scatter plots

- Additional categorical / numerical variables can be added using color, shape, size of dot ,...



Summary on visualization

Single variable plot plot counts

	type of plot
continuous variable	histogram
categorical variable	barplot

Two variable plot plot relationship

	continuous variable	categorical data
continuous variable	scatter plot	boxplot
categorical variable		heatmap

2. Describing data - mean/media/variance

Descriptive statistics

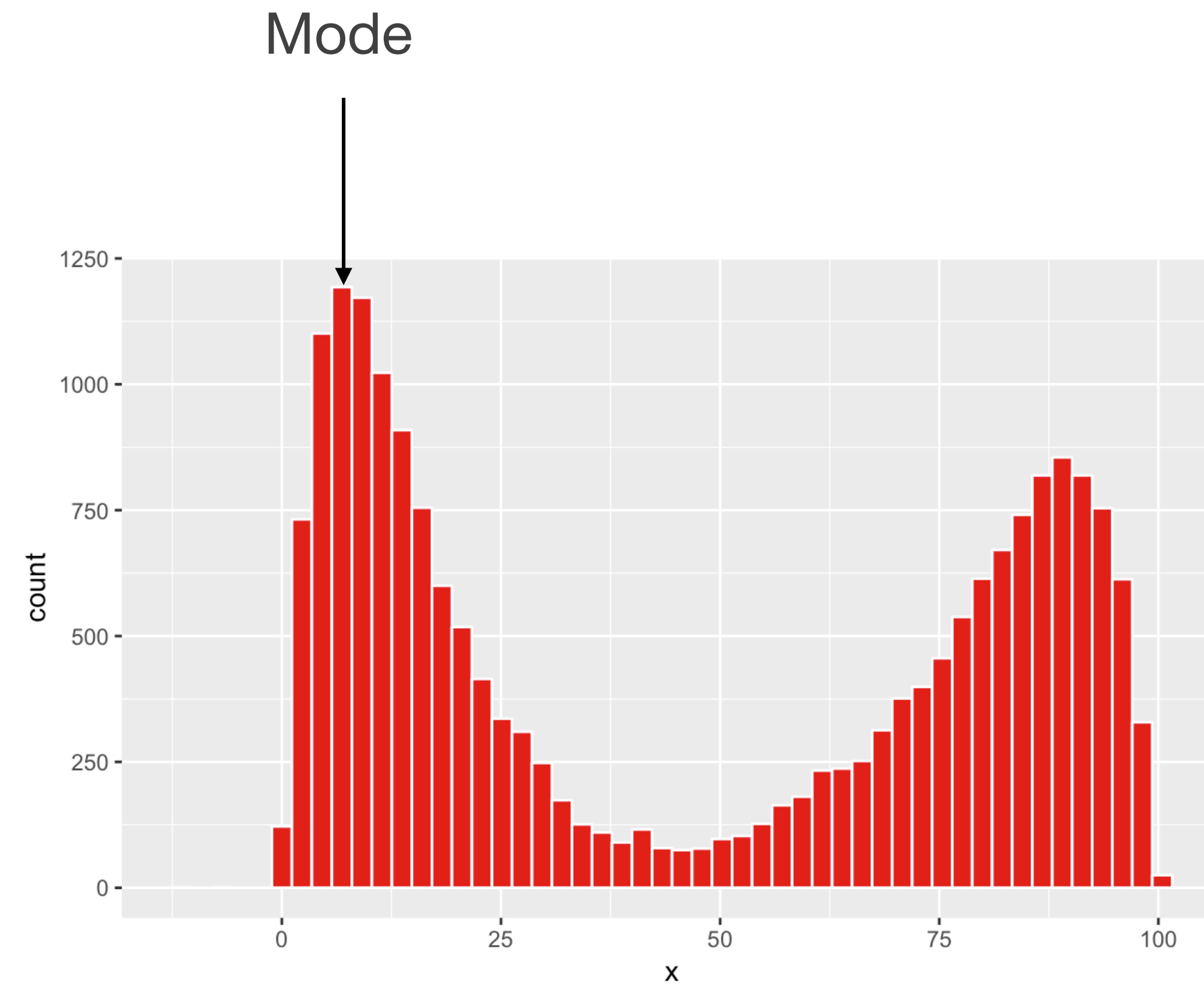
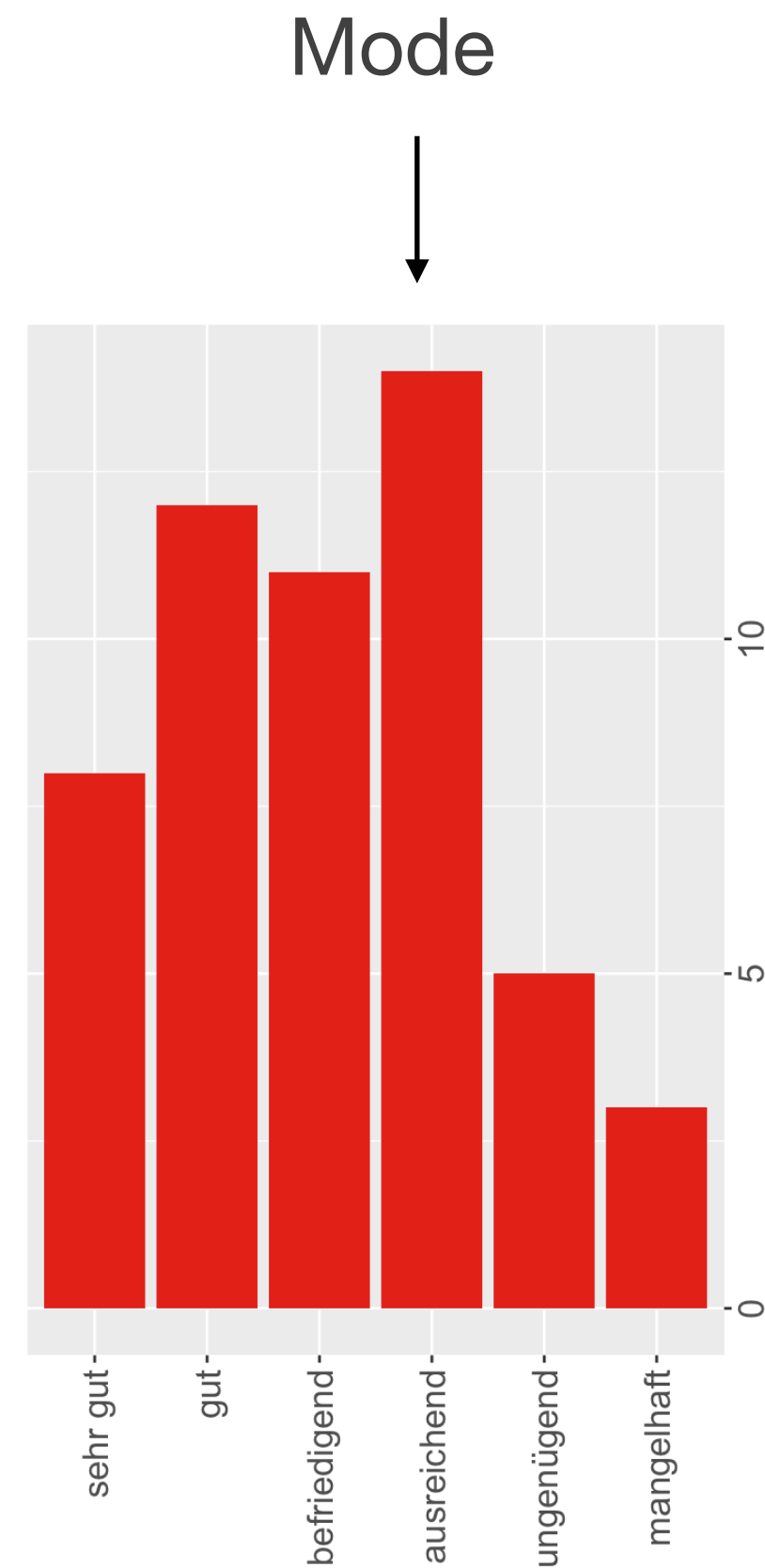
- Datasets represent samples taken from a larger population
- Example
 - diabetes dataset: sample representing a group of diabetes patients
 - gene expression dataset: samples (e.g. 10 mice before and after treatment) represent a group of mice extracted from a larger population
 - opinion polls: opinion of 1000 people extracted from a more general population

$$\{x_1, x_2, \dots, x_N\}$$

Descriptive statistics provides a description of the sample dataset,
without making any extrapolation to the more general population!

Computing the average

- Categorical data: **mode** = category with the highest count



Such a distribution is
called a ***“bimodal distribution”***

Computing the average

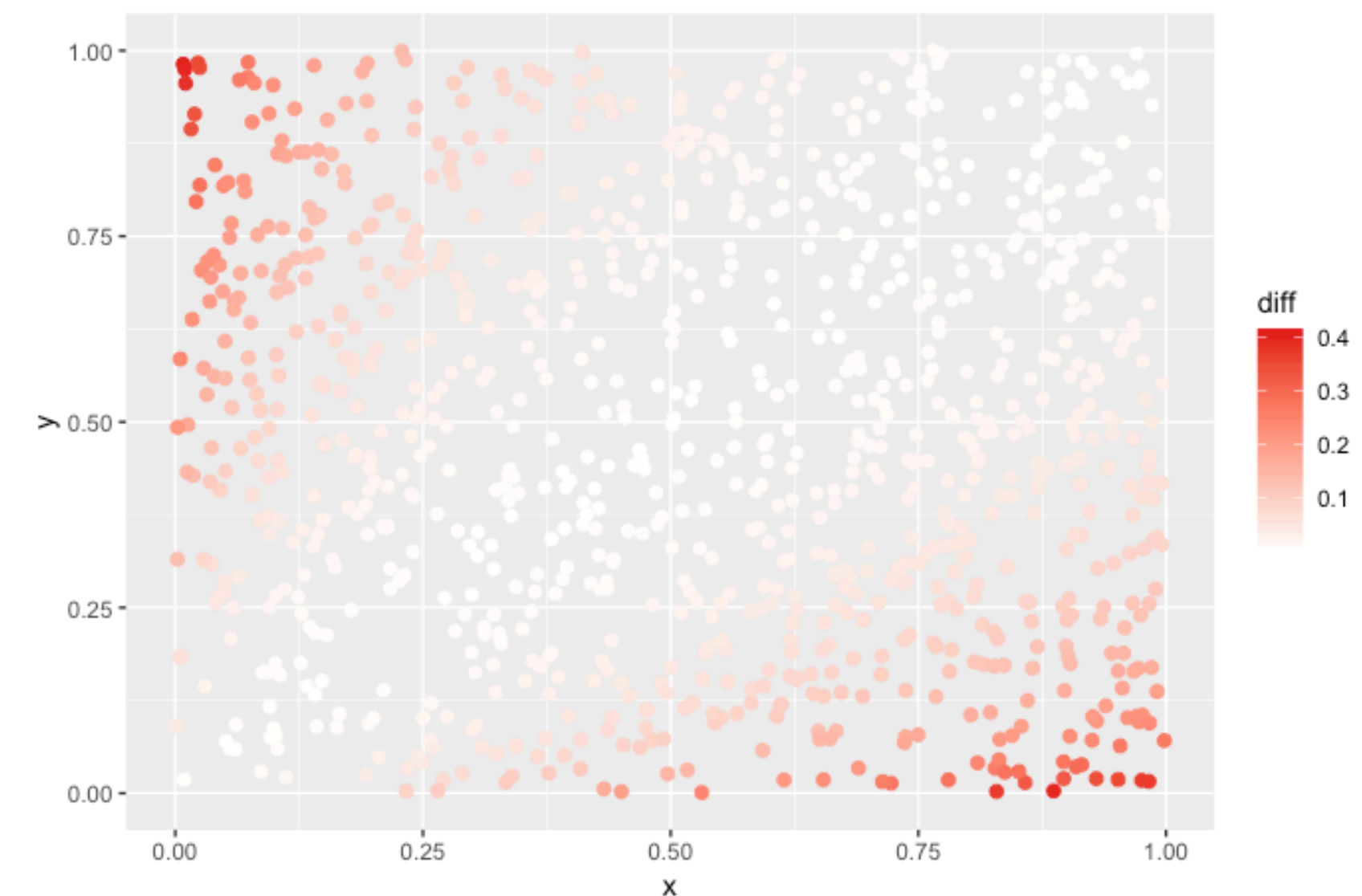
- Numerical data: **mean** of the data

$$\text{arithmetic mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{geometric mean} = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

- arithmetic mean is always greater/equal the geometric mean

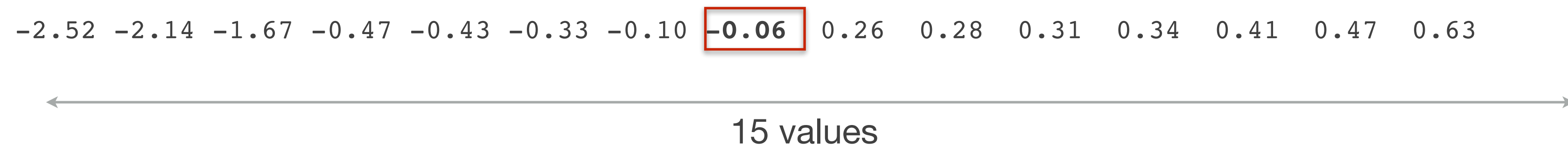
difference arithm. - geom.
for 2 values (x,y)



Computing the average

- Numerical data: **median**
- median = numerical value which splits the dataset into two equal parts
 - 50% of the values are larger, 50% of the values are smaller than the median value

- **Odd** number of values : sort and take the middle one

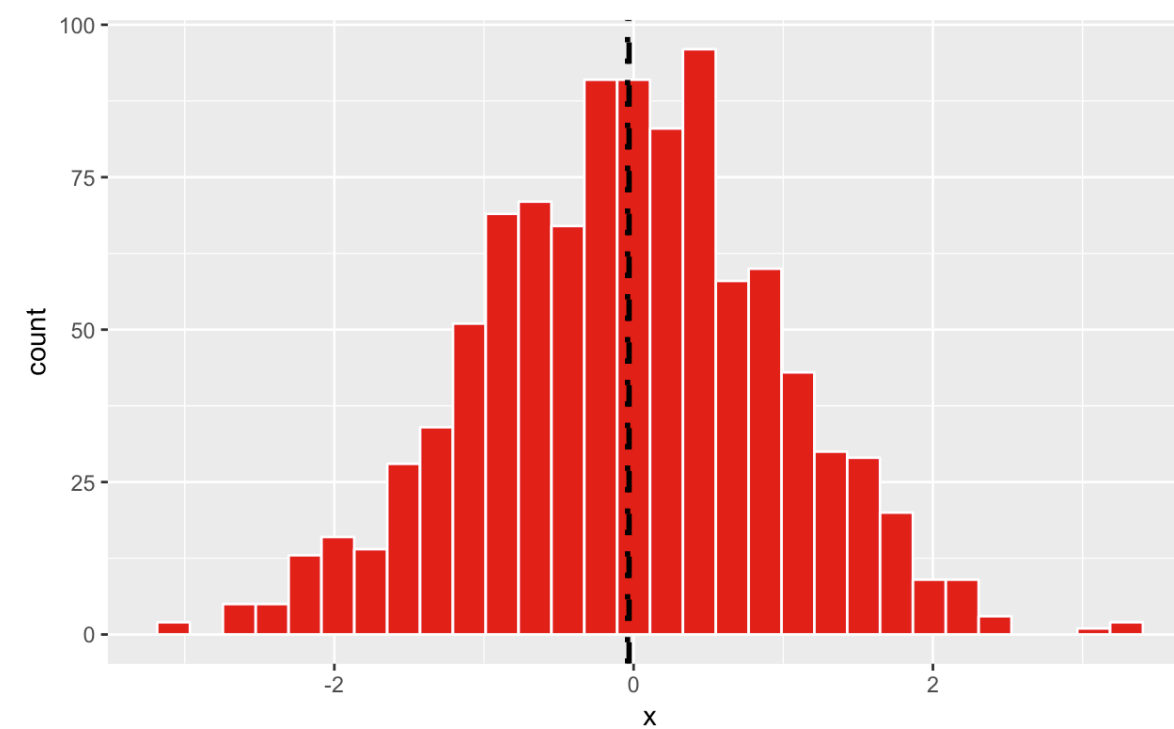


- **Even** number of values: sort and mean of the middle 2 values

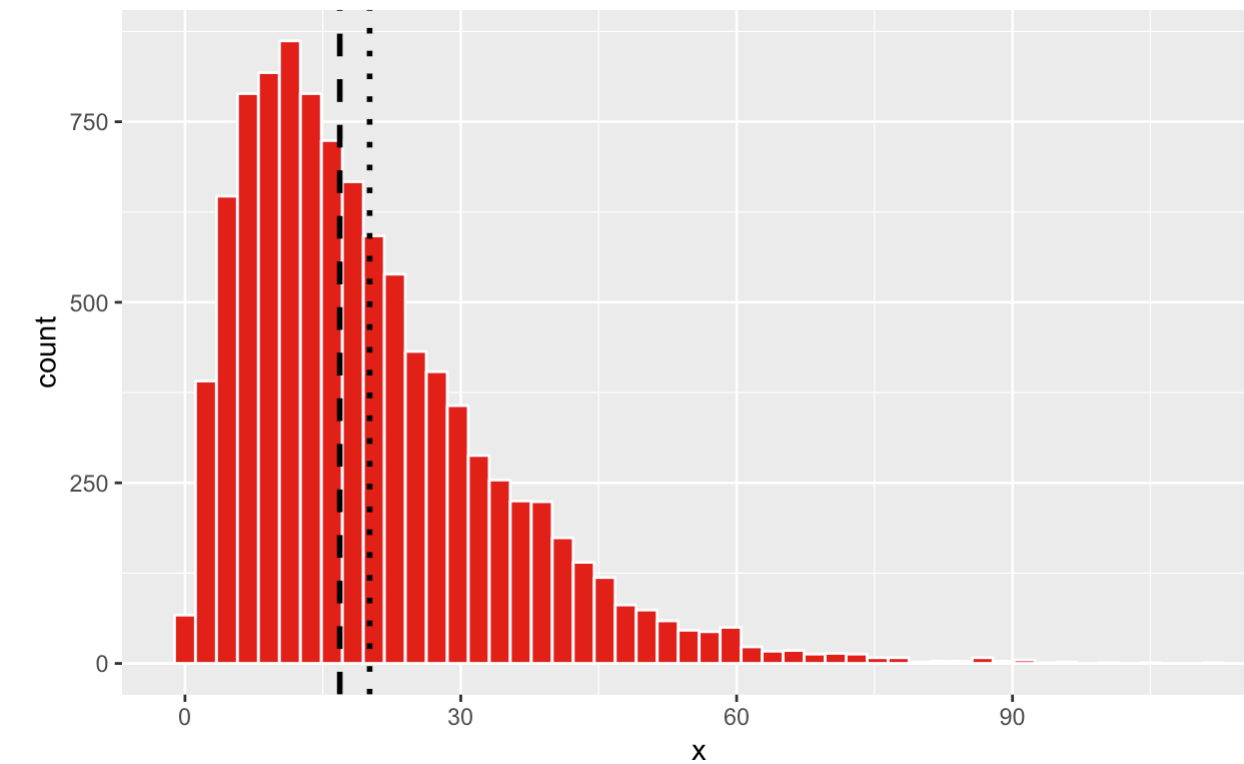


Mean / Median

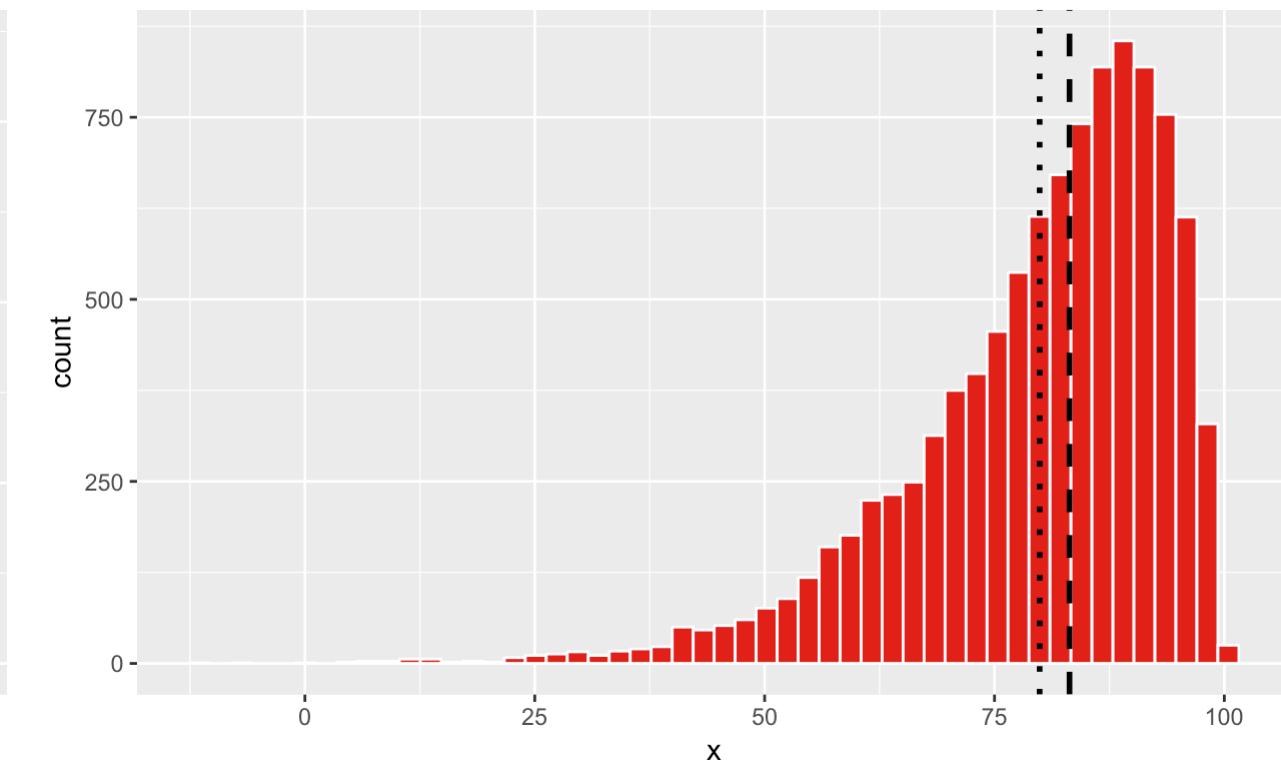
Which line is the median, which is the mean ?



symmetrical distribution
median ~ mean



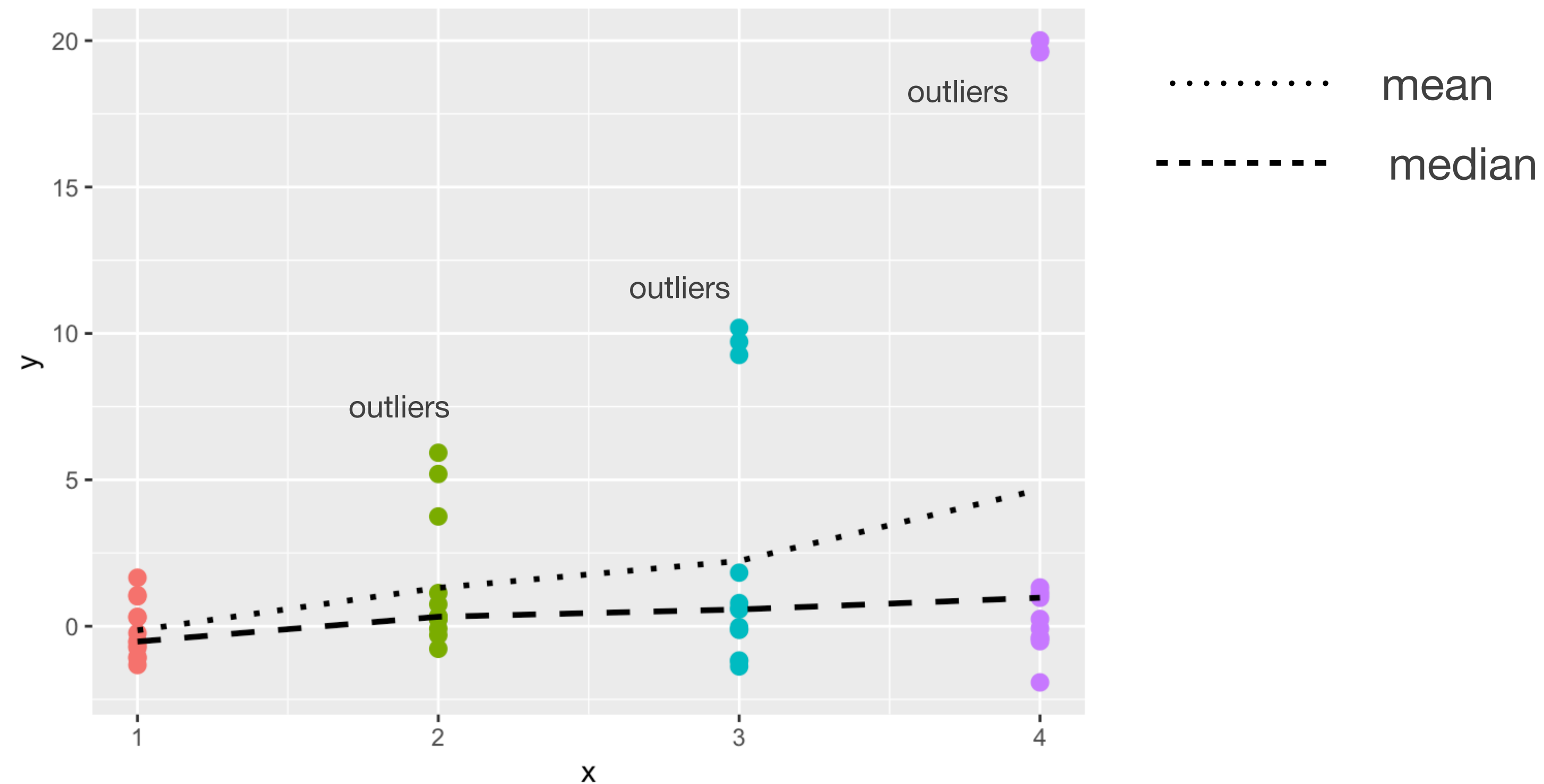
right-skewed distribution
mean > median



left-skewed distribution
mean < median

Mean vs. median

- 4 groups, with 3 outliers with increasing effect
- *median is more robust to outliers than mean*



Measuring data spread

- How much spread is there in the dataset?

- **Sample variance :**

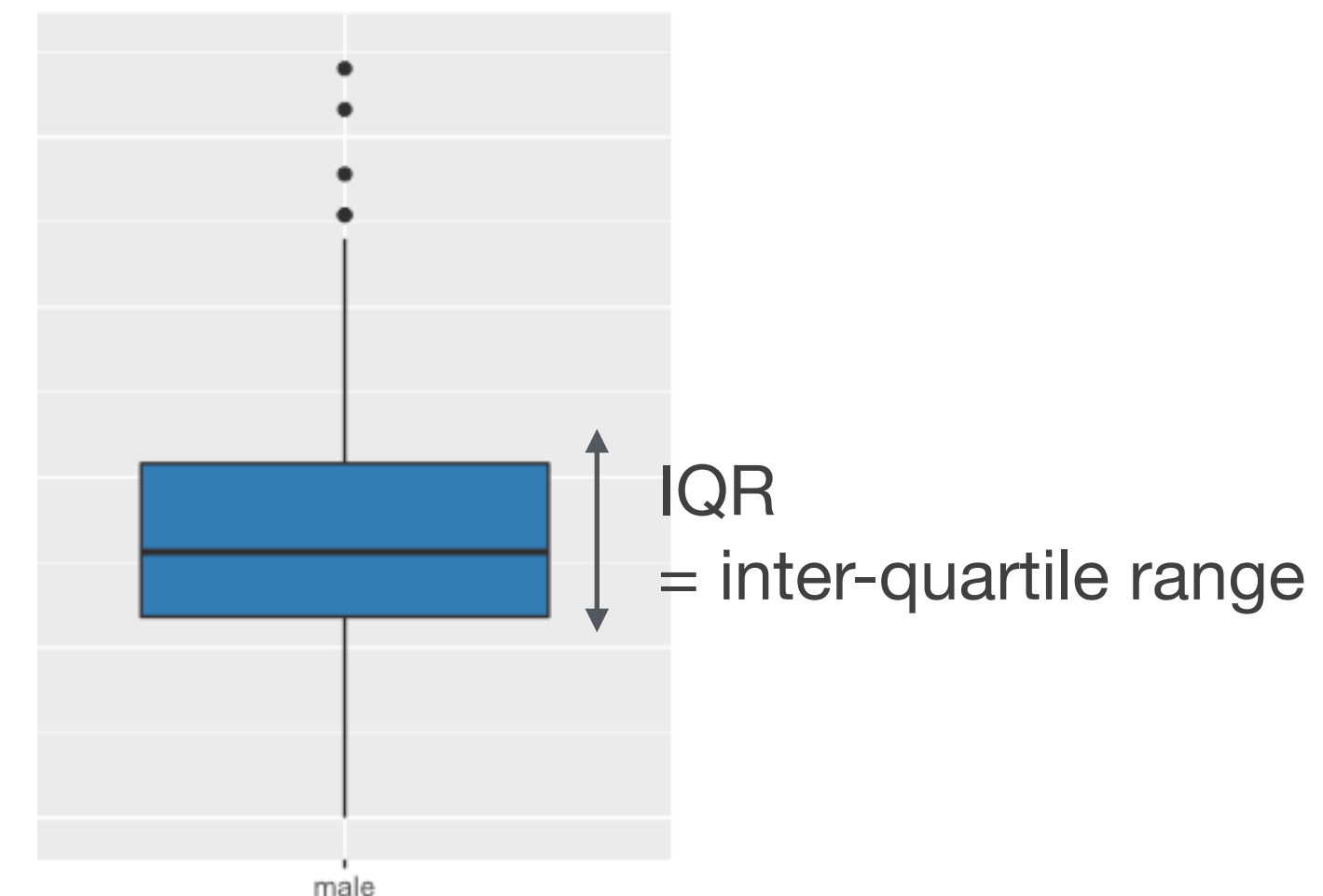
$$Var(x) = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

$\bar{x}$ = mean

- **Standard deviation :**

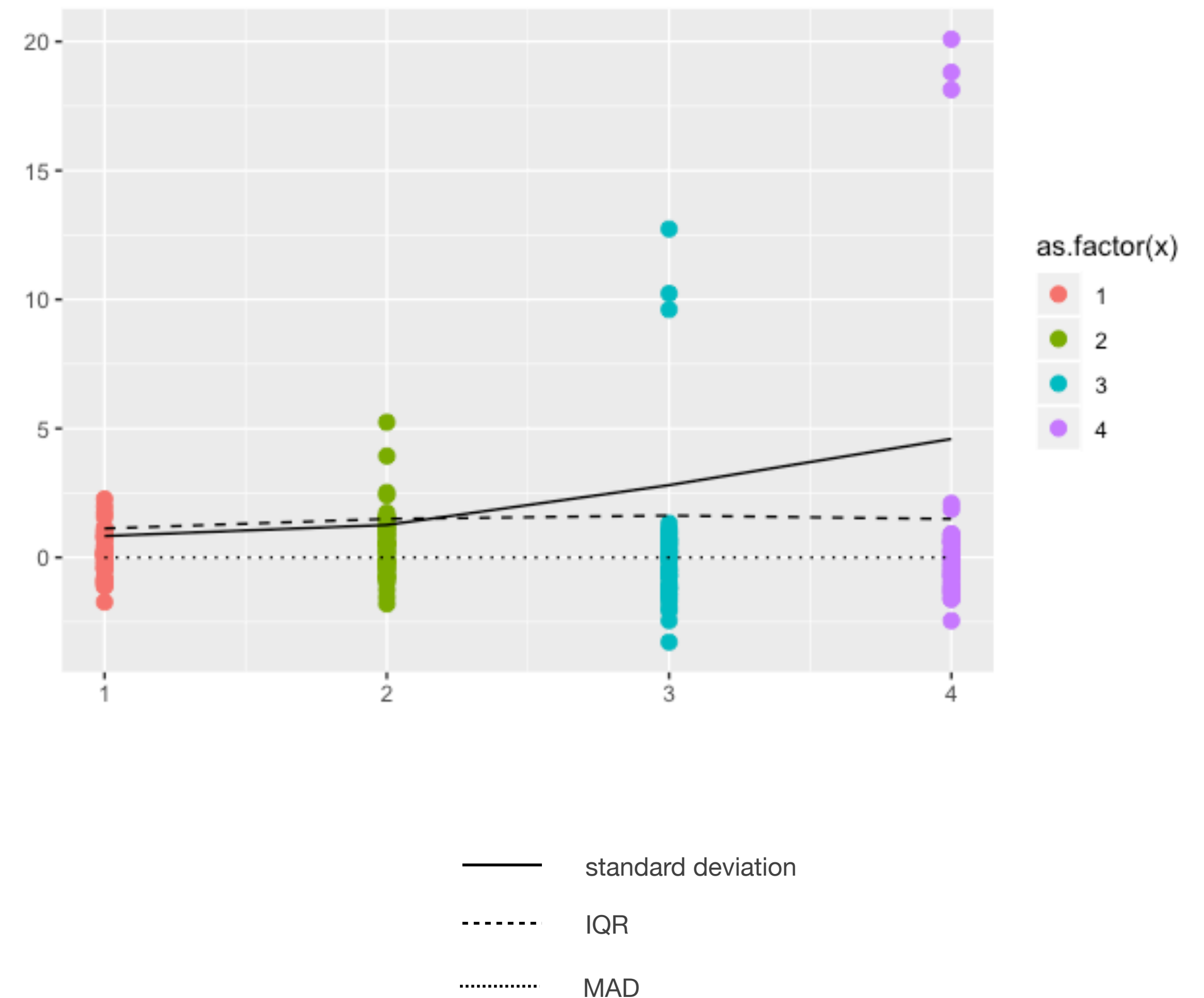
$$s_x = \sqrt{Var(x)}$$

- **Interquartile Range (IQR) :** difference between the 75% and 25% percentile



Measuring data spread

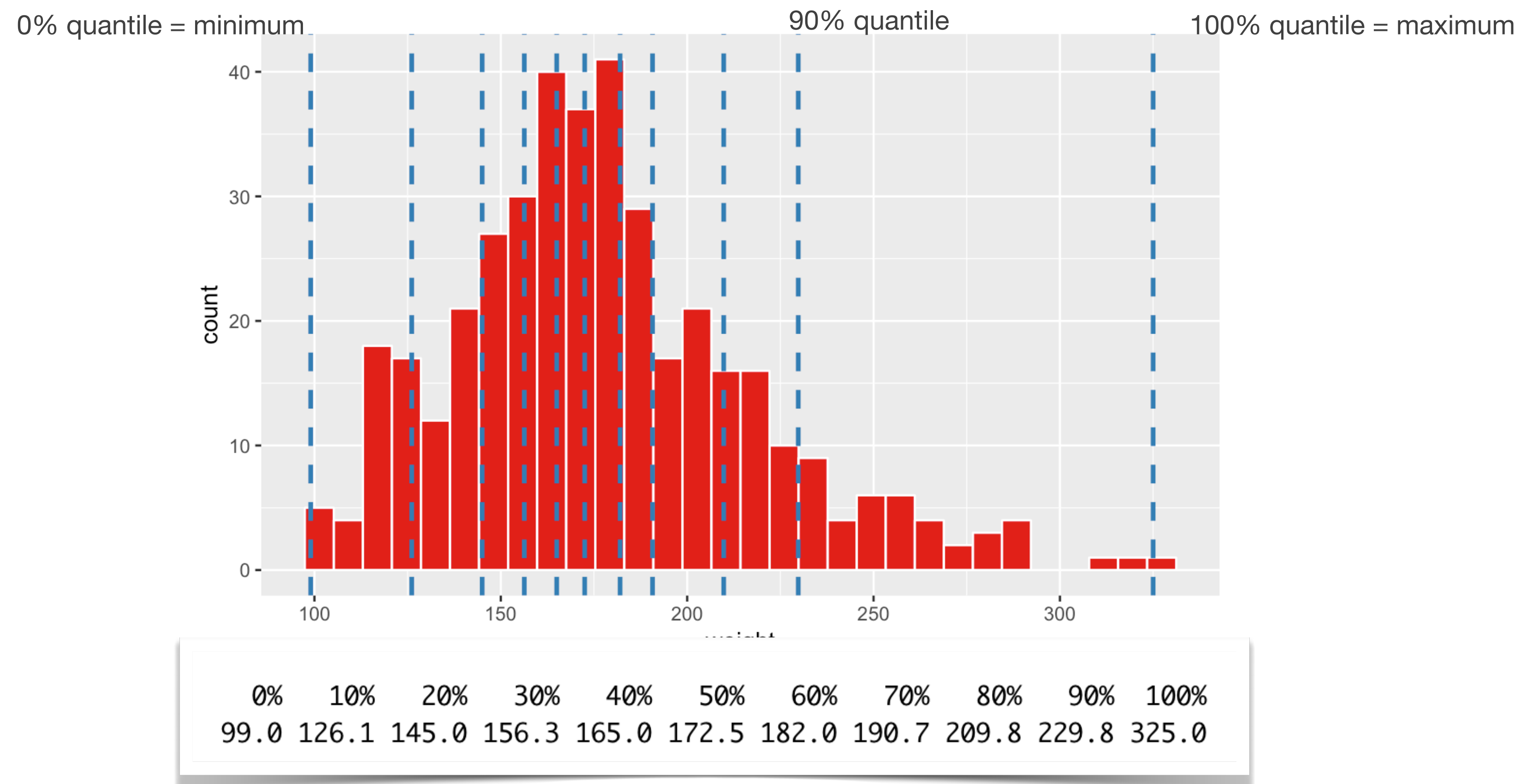
- **Median absolute deviation (MAD)**
median absolute deviation of the data points
from the median...
- How to compute?
 1. compute the median $med(x)$ of the dataset
 2. compute the absolute deviation of all
points from the median
 $d_i = |x_i - med(x)|$
 3. compute the median of the d_i
- IQR and MAD are less sensitive to outliers than
standard deviation !



2. Describing data - Quantiles / QQ-plots

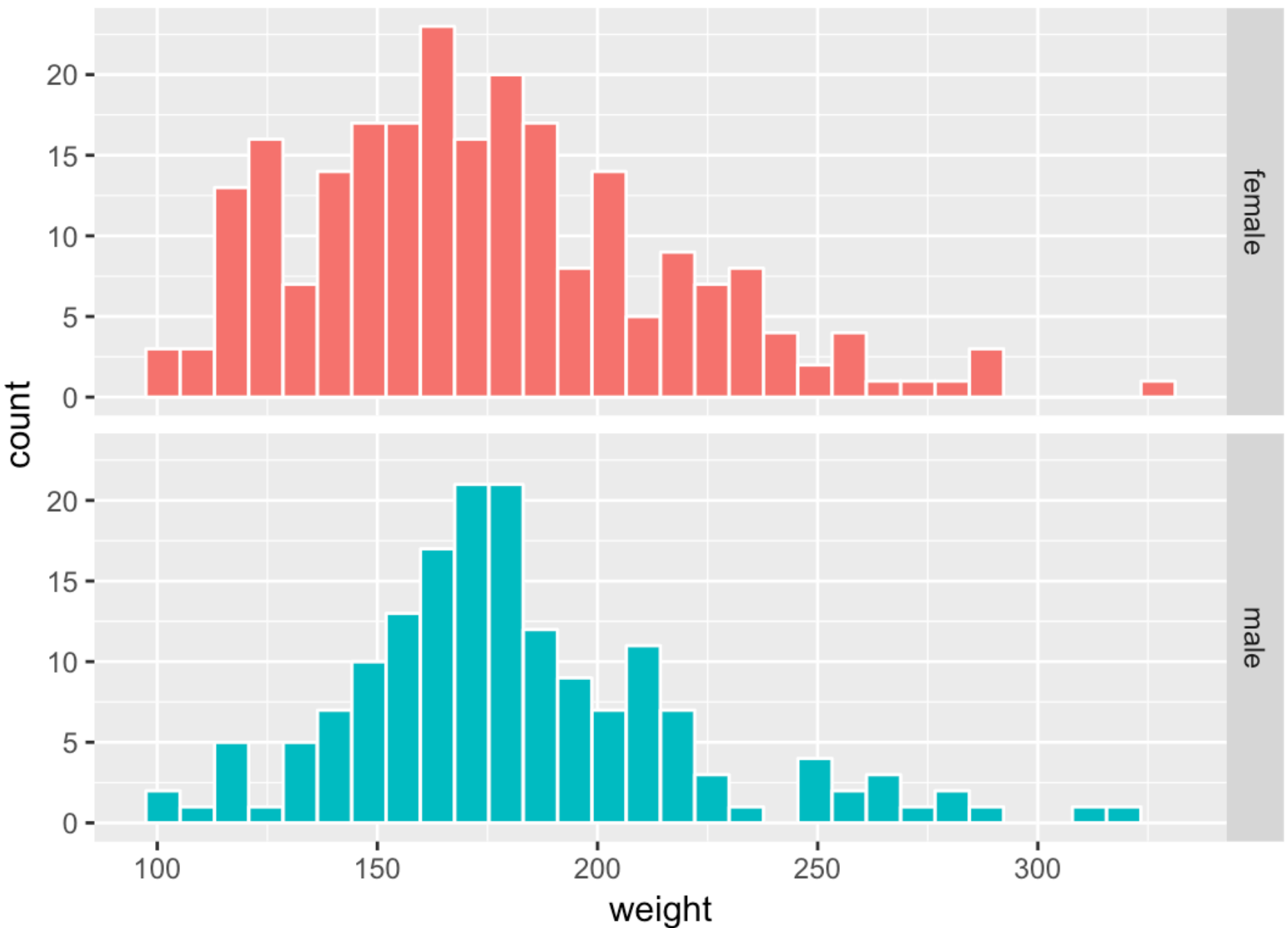
Quantiles

- **p-quantile** = value such that p% of the values in the data set are smaller
- distribution of quantile value given an indication about the **shape** of the distribution

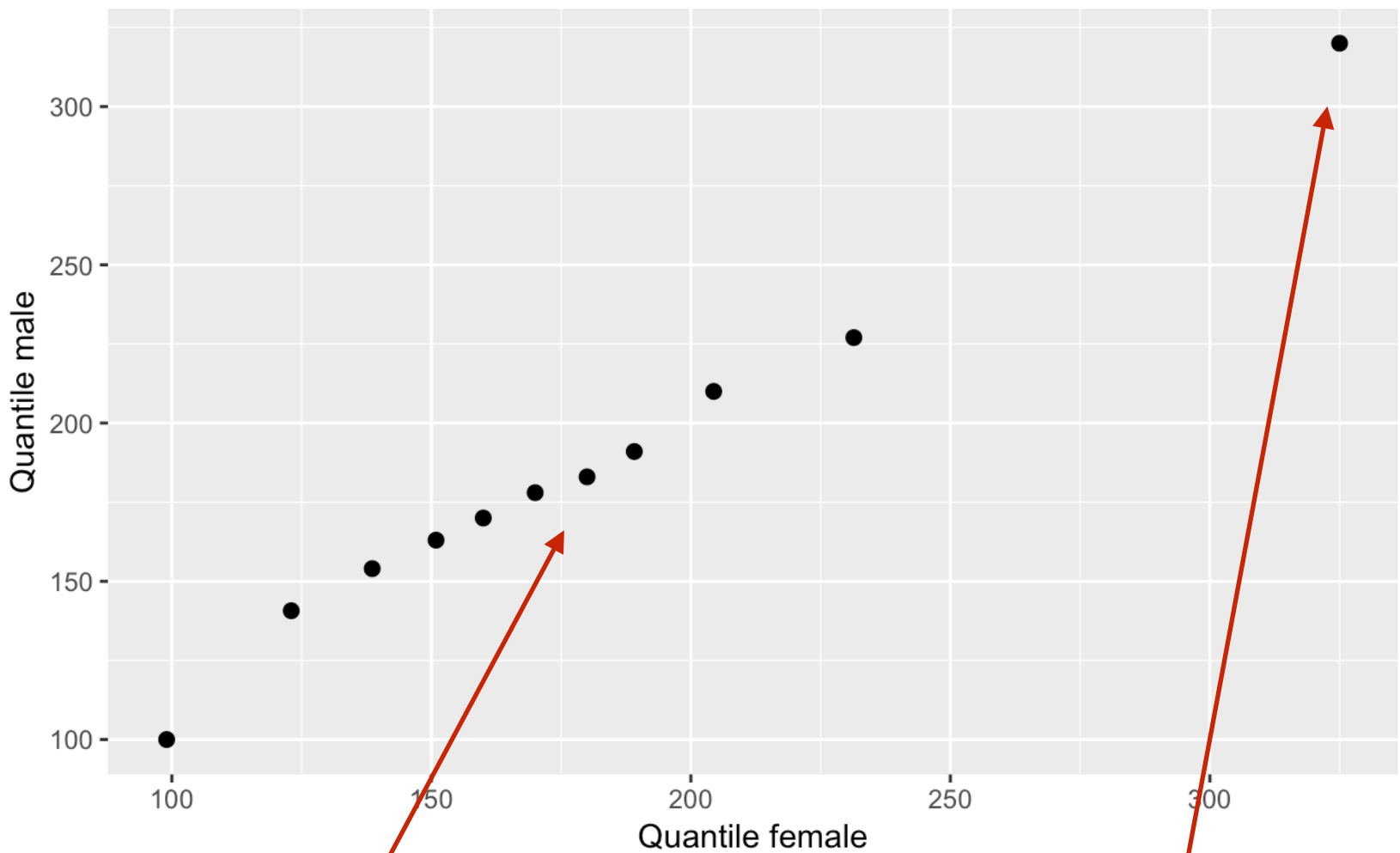


Quantile-quantile plots

weight distribution men / women



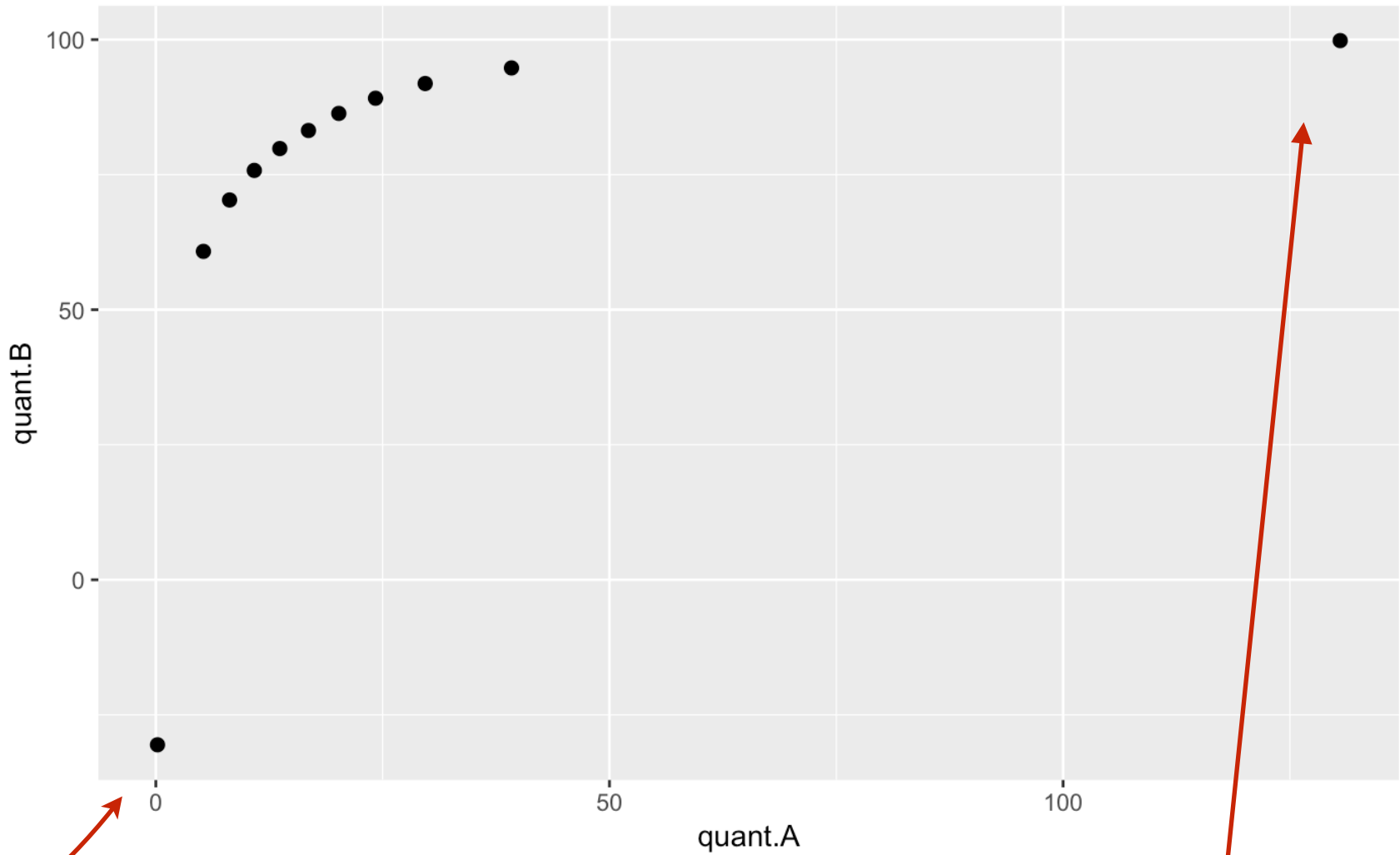
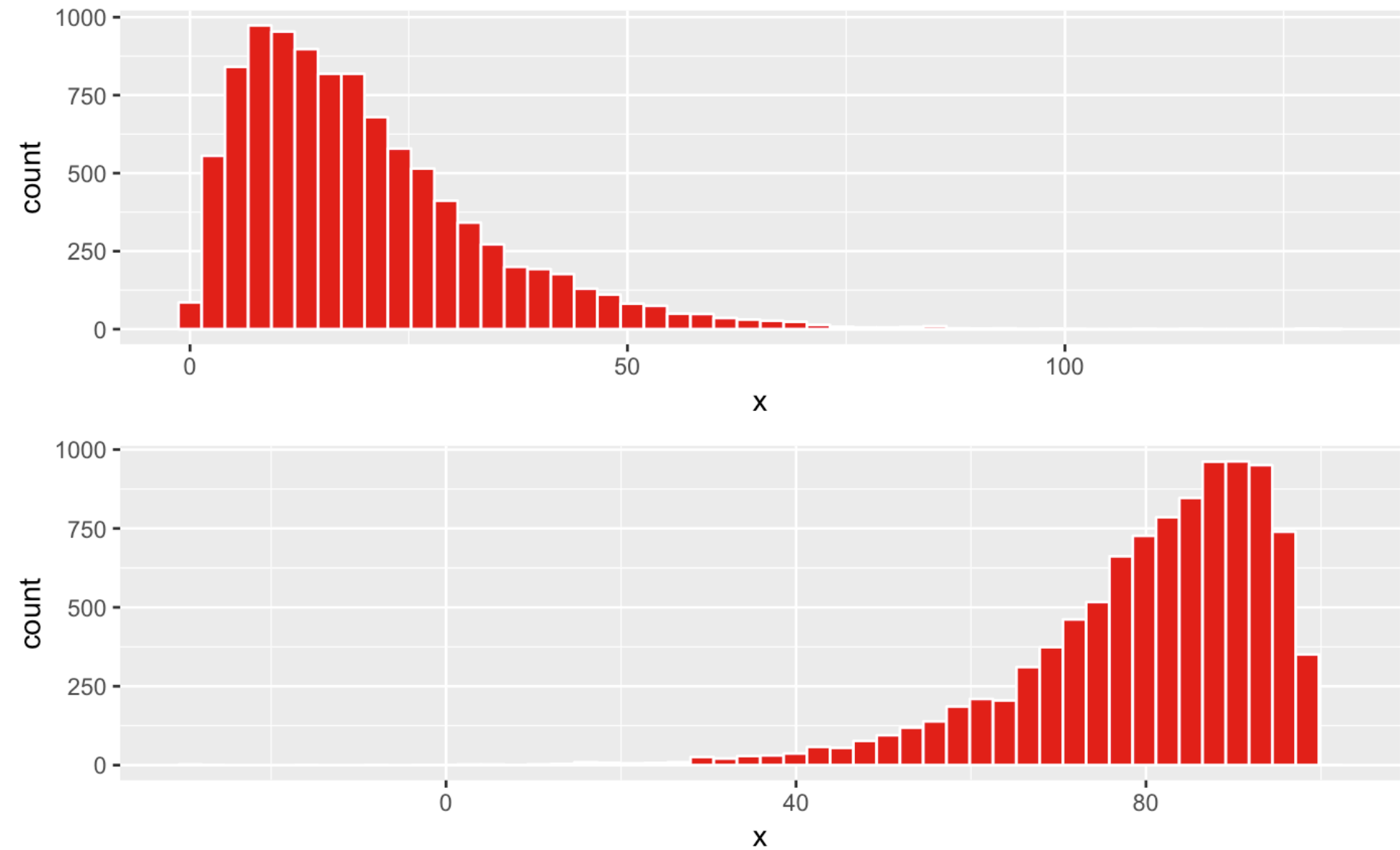
quantile-quantile plot



	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
quant.female	99	123.0	138.6	150.9	160	170	180	189.1	204.4	231.4	325
quant.male	100	140.7	154.0	163.0	170	178	183	191.0	210.0	227.0	320

if QQ-plot is a straight line, then both distributions
have a similar shape (up to translations / dilatation)

Quantile-quantile plots



	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
quant.A	0.18	5.24	8.13	10.85	13.66	16.82	20.16	24.22	29.69	39.20	130.55
quant.B	-30.55	60.80	70.31	75.78	79.84	83.18	86.34	89.15	91.87	94.76	99.82

Week 1 - Stop and think !

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

1. Which measure of spread is MOST robust to outliers?
 - A. Standard deviation
 - B. The spread between 10% and 90% quantile
 - C. Median
 - D. Median Absolute Deviation (MAD)
 - E. Range
2. For a right skewed-distribution, what is the order between mode, median and mean?
3. Can you imagine a distribution for which the mean is smaller than the mode?
4. Do you think that the mean and the variance are related? Think for example about the expression of a gene across multiple samples (slide 5)
5. What information is missing in Boxplots? How could one improve this?